Combinatorics 2019 Fall

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Textbooks (recommended):

1.《组合数学》潘永亮, 徐俊明, 科学出版社, 2006.

2. Extremal Combinatorics, with applications in Computer Science. 2nd Edition. Stasys Jukna, Springer, 2011.

1 Sep 3: bijection, binomial coefficient, double counting

Notation:

(1) A finite set X is a collection of distinct elements. The size of X is denoted by $|X|$, where $|X| :=\#$ (the number of) elements of X.

(2) for an integer $n > 0$, let $[n] := \{1, 2, \cdots, n\}$.

(3) A <u>vector (or string) of length n over X</u> is written as $(x_1, \cdots, x_n)$, where $x_i \in X$ for $i \in [n]$. The total number of vectors of length n over X is $|X|^n$.

(4) Some commonly used abbreviations:
"i.e." means "that is"; "$\exists$" means "exist"; "$\Rightarrow$" means "imply"; "s.t." means "such that"; "!" means "unique"; "$\forall$" means "for all".

Bijections:

Definition. *For two sets A, B, a <u>function</u> f: $A \to B$ is a relation between A and B, $(a, f(a))$, s.t. for each $a \in A$, there is exactly one $b \in B$ satisfying $b = f(a)$.*

Definition. *An <u>injection</u> (or one to one function) is a function s.t. if $a \neq a'$, then $f(a) \neq f(a')$; A <u>surjection</u> (or onto function) is a function satisfying $\forall\ b \in B,\ \exists\ a \in A,$ s.t. $f(a) = b$; if a function is both injection and surjection, then we say it is a <u>bijection</u>.*

Proposition 1.1. *For two finite sets A, B, $f : A \to B$ is a function.*

(1) if f is an injection, then $|A| \leq |B|$;

(2) if f is a surjection, then $|A| \geq |B|$;

(3) if f is a bijection, then $|A| = |B|$.

Proof. (1) Proof by contradiction. Assume that $|B| < |A|$. Since f is a function, by definition, $\forall\ a \in A,\ \exists!\ b \in B$ s.t.$f(a) = b$. Since $|B| < |A|$, then $\exists\ a \neq a'$ s.t. $f(a) = f(a')$ (by <u>pigeonhole principal</u>). This is a contradiction with the fact that f is an injection.

(2) Proof by contradiction. Assume that $|B| > |A|$. Since f is a function, by definition, $\forall\ a \in A,\ \exists!\ b \in B$ s.t.$f(a) = b$. Since $|B| > |A|$, $\exists\ b \in B$ s.t. there does not exist $a \in A$ satisfying $f(a) = b$. This is a contradiction with the fact that f is a surjection.

(3) (1)&(2) $\Rightarrow$ (3).

□

Proposition 1.2. *For two finite sets X, Y, with $|X| = n$ and $|Y| = r$. Let $X^Y := \{all\ functions\ f : Y \to X\}$. Then $|X^Y| = n^r$.*

Proof. Let B={all vectors of length r over X}, then $|B| = n^r$. Assume that $Y = \{y_1, y_2, \cdots, y_r\}$. Define a function

$$g : X^Y \to B.$$

$$f \mapsto (f(y_1), \cdots, f(y_r))$$

(1) g is a well-defined function, i.e, for each $f \in X^Y$, there is a unique vector $(f(y_1), \cdots, f(y_r)) \in B$ related to f.

(2) g is an injection, since $f \neq f' \Rightarrow (f(y_1), \cdots, f(y_r)) \neq (f'(y_1), \cdots, f'(y_r))$.

(3) we claim that g is a surjection. In fact, $\forall\ (b_1, \cdots, b_r) \in B$, define a function $f : Y \to X$ by $f(y_i) = b_i$, $i \in [r]$. It is easy to check that f is a well-defined function in X^Y. By the definition of g, we have $g(f) = (b_1, \cdots, b_r)$.

Combining all pieces above, we have that g is a bijection. By Proposition 1.1, $|X^Y| = |B| = n^r$. □

Proposition 1.3. *Let $Y = [r]$, denote 2^Y :={all subsets of Y}. Then $|2^Y| = 2^r$.*

Proof. Let B ={all vectors of length r over {0,1}}. Define a function $f : 2^Y \to B$ by $A \to f(A) = (b_1, \cdots, b_r)$, where

$$b_j = \begin{cases} 0, & \text{if } i \notin A, \\ 1, & \text{if } i \in A. \end{cases}$$

<u>Exercise:</u> show that f is a bijection, and then $|2^Y| = |B| = 2^r$. □

Remark. *In the proof of Proposition 1.3, f is called the indicator (or characteristic) function, $f(A)$ is called the <u>indicator vector</u> of A in Y, while A is called the <u>support</u> of $f(A) = (b_1, \cdots, b_r)$.*

Proposition 1.4. *Let $|X| = n$, $|Y| = r$, with $n \geq r$, then the number of injections $f : Y \to X$ equals $n(n-1)\cdots(n-r+1) := (n)_r$, where $(n)_r$ is called the r-th factorial of n.*

Proof. Let $A = \{$all injections $f : Y \to X\}$, $B = \{$all vectors of length r over X with r different entries$\}$.

<u>Exercise:</u> define a bijection $g : A \to B$ and complete the proof. $\square$

Binomial Coefficient:

- Let $|X| = n$, denote $\binom{X}{k} := \{A \subset X : |A| = k\}$.
- Let $\binom{n}{k} := \#k-$subsets of an $n-$elements set, i.e., $\left|\binom{X}{k}\right| = \binom{n}{k}$. Then $\binom{n}{0} = 1$, and $\binom{n}{k} = 0$ if $k > n$.
- Denote the factorial of n by $n! := n(n-1)\cdots 2 \cdot 1$. Define $0! = 1$.

Proposition 1.5. $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

Proof. Let $B = \{$all vectors of length k over $[n]$ consisting k different elements$\}$. We use the method of <u>Double Counting</u> to prove the equality.

Way 1: Directly count the number of vectors, $|B| = n(n-1)\ldots(n-k+1) = \frac{n!}{(n-k)!}$.

Way 2: There are $\binom{n}{k}$ ways to choose a $k-$subset of X. For each $k-$subset, there are $k!$ ways to order it to a vector. So $|B| = \binom{n}{k}k!$.

Combining Way 1 and 2, we have $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. $\square$. $\square$

2 Sep 5: bijection, combinatorial method

Proposition 2.1. *(1)* $\binom{n}{k} = \binom{n}{n-k}$.

(2) (<u>Pascal Triangle</u>) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof. (1) $|X| = n$. Define $f : \binom{X}{k} \to \binom{X}{n-k}$, $f(A) = X \backslash A$. It's easy to check that f is a bijection.

(2) $|X| = n$. Separate $\binom{X}{k}$ into two parts, and fix an element $t \in X$.

- $\#\{\text{all } k\text{−subsets containing } t\}=\binom{n-1}{k-1}$.
- $\#\{\text{all } k\text{−subsets avoiding } t\}=\binom{n-1}{k}$.

Combine these two situations, and we prove (2).

□

Selections with repetition: select k balls from n boxes

Proposition 2.2. *# integer solutions to $x_1 + \cdots + x_n = k$ where $x_i > 0$ is $\binom{k-1}{n-1}$.*

Proof. This question is equivalent to ask: *How many ways are there of distributing k sweets to n children such that each child has at least one sweet.*

Lay out the sweets in a single row of length k, and cut it into n pieces. Then give the sweets of the i-th piece to child i, which means that we need $n-1$ cuts from $k-1$ possibles.

$$\Rightarrow \binom{k-1}{n-1}.$$

□

Proposition 2.3. *# integer solutions to $x_1 + \cdots + x_n = k$ where $x_i \geqslant 0$ is $\binom{n+k-1}{n-1}$.*

Proof. Let $A = \{\text{integer solutions to } x_1 + \cdots + x_n = k, x_i \geq 0\}$ and $B = \{\text{integer solutions to } y_1 + \cdots + y_n = n + k, y_i > 0\}$. Then $|B| = \binom{n+k-1}{n-1}$ by Proposition 2.2.

Define $f : A \to B$, by $f((x_1, \cdots, x_n)) = (x_1 + 1, x_2 + 1, \cdots, x_n + 1)$. It suffices to check that <u>f is a bijection</u>.

(1) f is well-defined: For any $(x_1, \cdots, x_n) \in A$, $\exists!\ (y_1, \cdots, y_n) = (x_1 + 1, x_2 + 1, \cdots, x_n + 1) \in B$ s.t. $f((x_1, \cdots, x_n)) = (y_1, y_2, \cdots, y_n)$.

(2) f is injective.

(3) f is surjective.

□

Proposition 2.4. *$X = [n]$, $A = \{(a_1, \cdots, a_r) :\ a_i \in X,\ 1 \leqslant a_1 \leqslant a_2 \leqslant \cdots \leqslant a_r \leqslant n,\ a_{i+1} - a_i \geqslant k + 1,\ i \in [r-1]\}$. Then $|A| = \binom{n-k(r-1)}{r}$.*

Proof. Let $B = \{(b_1, \cdots, b_{r+1}) : b_1 + \cdots + b_{r+1} = n-1,\ b_1 \geqslant 0,\ b_i \geqslant k+1,\ i = 2, \cdots, r,\ b_{r+1} \geqslant 0\}$. Define $f : A \to B$, $(a_1, \cdots, a_r) \mapsto (b_1, \cdots, b_{r+1})$ by

$$b_1 = a_1 - 1 \geqslant 0.$$

$$b_i = a_i - a_{i-1} \geqslant k+1, i = 2, \cdots, r.$$

$$b_{r+1} = n - a_r \geqslant 0.$$

It's easy to check that f is a bijection. Then $|A| = |B|$

Now, let $C = \{(c_1, \cdots, c_{r+1}) :\ c_1 + \cdots + c_{r+1} = n - 1 - (k+1)(r-1),\ c_1, \cdots, c_{r+1} \geqslant 0\}$. Define $g : B \to C$, $(b_1, \cdots, b_{r+1}) \mapsto (c_1, \cdots, c_{r+1})$ by

$$c_1 = b_1 \geqslant 0.$$

$$c_i = b_i - (k+1) \geqslant 0, i = 2, \cdots, r.$$

$$c_{r+1} = b_{r+1} \geqslant 0.$$

Also, it's easy to check that g is a bijection. Hence,

$$|B| = |C| = \binom{n-1-(k+1)(r-1)+(r+1)-1}{(r+1)-1} = \binom{n-k(r-1)}{r}. \square$$

$\square$

Arrangements with repetition

Proposition 2.5. *$X = \{x_1, \cdots, x_n\}$, $B =\{$all vectors of length r over X such that x_i occurs a_i times, $i \in [n]$, $\sum_{i=1}^{n} a_i = r\}$. Then,*

$$|B| = \frac{r!}{a_1! a_2! \cdots a_n!}.$$

Proof. trivial. $\square$

Corollary 2.6. *(Multinomial Theorem)*

$$(x_1 + x_2 + \cdots + x_n)^r = \sum_{a_1 + a_2 + \cdots + a_n = r} \frac{r!}{a_1! a_2! \cdots a_n!} {x_1}^{a_1} \cdots {x_n}^{a_n}.$$

Proof.

$$(x_1 + x_2 + \cdots + x_n)^r = \sum_{1 \leqslant i_1, i_2, \cdots, i_r \leqslant n} x_{i_1} x_{i_2} \cdots x_{i_r}$$

The coefficient of ${x_1}^{a_1} \cdots {x_n}^{a_n}$ is equal to the number of vectors $(i_1, \cdots, i_n)$ over $[n]$ such that i occurs a_i times. By Proposition 5, we get this theorem. $\square$

Remark. *(1)* ***(Binomial Theorem)*** *If $n = 2$, $x_1 = a$, $x_2 = b$, then we get* $(a+b)^r = \sum_{i=0}^{r} \binom{r}{i} a^i b^{r-i}$.

(2) If $a = b = 1$, then we get $2^r = \sum_{i=0}^{r} \binom{r}{i}$

<u>Partition:</u> $X = R_1 \cup R_2 \cup \cdots \cup R_n$. There are two cases:

- unordered partition $\{R_1, R_2, \cdots, R_n\} = \{R_2, R_1, \cdots, R_n\}$.
- ordered partition $(R_1, R_2, \cdots, R_n) \neq (R_2, R_1, \cdots, R_n)$.

Proposition 2.7. $|X| = r$, $A =\{$*ordered partitions of X into n parts such that the i-th part has size a_i, $i \in [n]$,* $\sum_{i=1}^{n} a_i = r\}$. *Then*

$$|A| = \frac{r!}{a_1! a_2! \cdots a_n!}.$$

Proof. Method 1: This question is equivalent to *How many ways are there of partitioning r students into class $1, \cdots, n$ such that class i has a_i students.* There are $\binom{r}{a_1}$ ways of choosing the students in class 1. Then, there are $\binom{r-a_1}{a_2}$ ways of choosing the students in class 2. $\cdots$. So, there are

$$\binom{r}{a_1}\binom{r-a_1}{a_2}\binom{r-a_1-a_2}{a_3}\cdots\binom{r-a_1-\cdots-a_{n-1}}{a_n}$$

ways to partition the students as required, which is exactly $\frac{r!}{a_1! a_2! \cdots a_n!}$.

Method 2: Let B be the set defined in Proposition 2.5. <u>Exercise:</u> define a bijection from A to B. □

Proposition 2.8. $|X| = r$, $S =\{$*unordered partitions of X such that there are k_i parts of size i, $i \in [r]$,* $\sum_{i=1}^{r} ik_i = r\}$. *Then,*

$$|S| = \frac{r!}{(1!)^{k_1}(2!)^{k_2}\cdots(r!)^{k_r} k_1! k_2! \cdots k_r!}.$$

Proof. This question is equivalent to *How many ways are there of partitioning r students into different unordered groups such that there are k_i groups having i students.*

First, there are $\binom{r}{k_1 \cdot 1}$ ways of choosing the students in groups of size 1. To partion these students into these k_1 unordered groups, there are $\frac{(k_1 \cdot 1)!}{(1!)^{k_1}(k_1!)}$ ways.

Similarly, for $i = 2, \cdots, r$, there are

$$\binom{r - k_1 \cdot 1 - \cdots - k_{i-1} \cdot (i-1)}{k_i \cdot i}$$

ways of choosing the students in groups of size i. To partion these students into these k_i unordered groups, there are $\frac{(k_i \cdot i)!}{(i!)^{k_i}(k_i!)}$ ways.

So, after simple calculations, we get the desired result. $\square$

Stirling Number of the 2nd kind

$S(n,k) = \#$ of partition of [n] into k unordered non-empty subsets.

Then $S(n,0) = 0$, $S(n,1) = S(n,n) = 1$, $S(n,2) = 2^{n-1} - 1$, $S(n,n-1) = \binom{n}{2}$. And

$$S(n,k) = \sum_{\sum_{i=1}^{n} k_i = k, \sum_{i=1}^{n} ik_i = n} \frac{n!}{(1!)^{k_1}(2!)^{k_2}\cdots(n!)^{k_n} k_1! k_2! \cdots k_n!}.$$

Theorem 2.9. *$S(n,k) = S(n-1,k-1) + kS(n-1,k)$.*

Proof. Split the ways of partitions into two cases:

(1) n is alone in a subset, there are $S(n-1,k-1)$ ways.

(2) n is not alone in a subset. Then partition $[n-1]$ into k unordered non-empty parts, for which there are $S(n-1,k)$ ways. Then for each such partition, to join n in any subset, there are k ways. So there are in total $kS(n-1,k)$ ways.

$\square$

$C_n[\mathbb{Z}] := \{$all polynomials of deg $\leq n$ over $\mathbb{C}\}$.

Two sets of basis of it:

I: $\{1, z, z^2, ..., z^n\}$,

II: $\{(z)_0, (z)_1, (z)_2, ..., (z)_n\}$

where $(z)_0 = 1$, $(z)_n = z(z-1)...(z-n+1)$, $n \geq 1$.

Represent I by II, $z^n = \sum_{k=0}^{n} F(n,k)(z)_k$.

Hence $F(n,0) = 0$, $n > 0$. $F(n,n) = 1$, $n \geq 0$. $F(n,k) = 0$, $k > n$.

Theorem 2.10. *$F(n,k) = F(n-1,k-1) + kF(n-1,k)$, for $n \geq 1, k \geq 1$.*

Proof. Since $F(n,0)=0$, $z^n=\sum_{k=1}^{n}F(n,k)(z)_k$, $n\geq 1$.

$$\begin{aligned}
z^n &= z^{n-1}\cdot z = z\cdot\sum_{k=0}^{n-1}F(n-1,k)(z)_k\\
&= (z-k)\sum_{k=0}^{n-1}F(n-1,k)(z)_k + k\sum_{k=0}^{n-1}F(n-1,k)(z)_k\\
&= \sum_{k=0}^{n-1}F(n-1,k)(z)_{k+1} + \sum_{k=1}^{n-1}kF(n-1,k)(z)_k\\
&= \sum_{k=1}^{n}F(n-1,k-1)(z)_k + \sum_{k=1}^{n}kF(n-1,k)(z)_k\\
&= \sum_{k=1}^{n}(F(n-1,k-1)+kF(n-1,k))(z)_k
\end{aligned}$$

$\Longrightarrow F(n,k)=F(n-1,k-1)+kF(n-1,k).$ □

Corollary 2.11. $S(n,k)=F(n,k)$.

Stirling number of the 1st kind $s(n,k)$ is defined by

$$(z)_n=\sum_{k=0}^{n}s(n,k)z^k.$$

And $s(n,0)=0$, $n>0$; $s(n,n)=1$, $n\geq 0$; $s(n,k)=0$, $k>n$.

Theorem 2.12. $s(n,k)=s(n-1,k-1)-(n-1)s(n-1,k)$.

Proof. $n\geq 1$, $(z)_n=\sum_{k=1}^{n}s(n,k)z^k$.

$$\begin{aligned}
(z)_n &= (z)_{n-1}(z-n+1) = \sum_{k=0}^{n-1}s(n-1,k)z^k(z-n+1)\\
&= \sum_{k=0}^{n-1}s(n-1,k)z^{k+1} - (n-1)\sum_{k=0}^{n-1}s(n-1,k)z^k\\
&= \sum_{k=1}^{n}s(n-1,k-1)z^k - (n-1)\sum_{k=1}^{n}s(n-1,k)z^k\\
&= \sum_{k=1}^{n}[s(n-1,k-1)-(n-1)s(n-1,k)]z^k
\end{aligned}$$

$\Longrightarrow s(n,k)=s(n-1,k-1)-(n-1)s(n-1,k).$ □

Let $s(n,k)=(-1)^{n-k}Q(n,k)$, $Q(n,k)\geq 0$. Then

$$Q(n,k)=Q(n-1,k-1)+(n-1)Q(n-1,k).$$

Theorem 2.13. *Suppose S_n is symmetric group of $[n]$. Then the number of permutations having exactly k cycles is $Q(n,k)$.*

Proof. Let $P(n,k)$ be the set of permutations having k cycles in S_n. Split $P(n,k)$ into two parts.

(1) n is alone in a cycle, i.e. permutations having (n) as a cycle, there are $|P(n-1,k-1)|$ such permutations.

(2) n is not alone in a cycle. There are $|P(n-1,k)|$ permutations in S_{n-1} having k cycles. For each such permutation, put n in any of its cycles. for which there $n-1$ ways. So $(n-1)|P(n-1,k-1)|$ ways to get a permutation in S_n.

By (1) and (2), we have $|P(n,k)| = |P(n-1,k-1)| + (n-1)|P(n-1,k)|$.
Since the initial conditions $|P(n,0)| = Q(n,0) = 0$, $n > 0$; $|P(n,n)| = Q(n,n) = 1$. We have $|P(n,k)| = Q(n,k)$. □

Theorem 2.14. *(Vandermonde's Identity)*

(1) $\binom{m+n}{r} = \sum_{i=0}^{r} \binom{n}{i}\binom{m}{r-i} = \sum_{i+j=r} \binom{n}{i}\binom{m}{j}$.

(2) $\binom{m+n}{r+m} = \sum_{i-j=r} \binom{n}{i}\binom{m}{j}$.

Proof. (1) Since $(1+x)^{m+n} = (1+x)^m(1+x)^n$, by Binomial Theorem,

$$\Rightarrow \sum_{r=0}^{m+n} \binom{m+n}{r} x^r = \sum_{i=0}^{n} \binom{n}{i} x^i \cdot \sum_{j=0}^{m} \binom{m}{j} x^j.$$

Computing the coefficients of x^r for both sides, we get

$$\binom{m+n}{r} = \sum_{i+j=r} \binom{n}{i}\binom{m}{j} = \sum_{i=0}^{r} \binom{n}{i}\binom{m}{r-i}.$$

(2)

$$\sum_{i-j=r} \binom{n}{i}\binom{m}{j} = \sum_{i-j=r} \binom{n}{i}\binom{m}{m-j} = \sum_{i+(m-j)=m+r} \binom{n}{i}\binom{m}{m-j} = \binom{m+n}{r+m}.$$

□

Excercise 2.15. *Prove the following identities.*

(1) $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.

(2) $\sum_{k=0}^{n} \binom{m}{k}\binom{n}{p+k} = \binom{n+m}{p+m}$.

(3) $\sum_{k=1}^{m} \binom{m}{k}\binom{n-1}{k-1} = \binom{n+m-1}{n}$.

(4) $\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m} \quad (n \geq k \geq m)$.

(5) $\sum_{k=0}^{n} \binom{n}{k}\binom{k}{m} = \binom{n}{m}2^{n-m}$.

3 Sep 10: estimate, IEP

Definition. *$f(n) = O(g(n))$ means $\exists$ constants n_0 and C such that for $\forall\ n \geqslant n_0$, the inequality $|f(n)| \leqslant C \cdot g(n)$ holds.*

Fact. Let $C,\ \alpha,\ \beta,\ a > 0$ be fixed real numbers. Then

(1) $n^\alpha = O(n^\beta)$ if $\beta \geqslant \alpha$;

(2) $n^C = O(a^n)$ if $a > 1$;

(3) $(\ln n)^C = O(n^\alpha)$ if $\alpha > 0$.

Definition. *(1) $f(n) = o(g(n)) \Leftrightarrow \lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$;*

(2) $f(n) = \Omega(g(n)) \Leftrightarrow g(n) = O(f(n))$;

(3) $f(n) = \Theta(g(n)) \Leftrightarrow f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$;

(4) $f(n) \sim g(n) \Leftrightarrow \lim_{n\to\infty} \frac{f(n)}{g(n)} = 1$.

Theorem 3.1. *For $\forall\ n \geqslant 1$, we have*

$$e(\frac{n}{e})^n \leqslant n! \leqslant en(\frac{n}{e})^n.$$

Proof. Consider $\int_1^n \ln x dx$, then

$$\ln(n-1)! = \sum_{i=1}^{n-1} \ln i \leqslant \int_1^n \ln x dx \leqslant \sum_{i=1}^{n} \ln i = \ln n!,$$

$$\Longrightarrow \ln(n-1)! \leqslant n\ln n - n + 1 \leqslant \ln n!.$$

Thus,

$$(n-1)! \leqslant e^{n\ln n - n + 1} \leqslant n!,$$

where $e^{n\ln n - n+1} = (e^{\ln n})^n e^{-n} e = (\frac{n}{e})^n e$. Therefore,

$$e(\frac{n}{e})^n \leqslant n! \leqslant en(\frac{n}{e})^n.$$

□

Excercise 3.2. *(1) Prove Theorem 3.1 by induction using the fact:* $1+x \leqslant e^x$.

(2) Prove a stronger result $n! \leqslant e\sqrt{n}(\frac{n}{e})^n$ *by definite integral.*

Stirling formula. $n! \sim \sqrt{2\pi n}(\frac{n}{e})^n$.

Fact. $\max\{\binom{n}{k} : k = 0, 1, 2, \cdots, n\} = \begin{cases} \binom{n}{\frac{n}{2}}, \text{if } n \text{ is even;} \\ \binom{n}{\lfloor \frac{n}{2} \rfloor} = \binom{n}{\lceil \frac{n}{2} \rceil}, \text{if } n \text{ is odd.} \end{cases}$

Corollary 3.3. $\frac{2^n}{n+1} \leqslant \binom{n}{\lfloor \frac{n}{2} \rfloor} \leqslant 2^n$.

Stirling approximation. $\binom{n}{\lfloor \frac{n}{2} \rfloor} \sim \frac{2^n}{\sqrt{n}}\sqrt{\frac{2}{\pi}}$.

Theorem 3.4. *For* $1 \leqslant k \leqslant n$*, we have*

$$(\frac{n}{k})^k \leqslant \binom{n}{k} \leqslant (\frac{en}{k})^k.$$

Proof. For lower bound, note that $\frac{n}{k} \leqslant \frac{n-i}{k-i}$ for $\forall\ i < k$, then

$$(\frac{n}{k})^k \leqslant \frac{n}{k} \cdot \frac{n-1}{k-1} \cdot \cdots \cdot \frac{n-k+1}{1} = \binom{n}{k}.$$

For upper bound, note that for any t s.t. $0 < t < 1$, we have

$$\binom{n}{k} \leqslant \sum_{i=0}^{k} \binom{n}{i} \leqslant \sum_{i=0}^{k} \binom{n}{i} \frac{t^i}{t^k} \leqslant \sum_{i=0}^{n} \binom{n}{i} \frac{t^i}{t^k} = \frac{(1+t)^n}{t^k}.$$

In particular, let $t = \frac{k}{n} < 1$, then

$$\binom{n}{k} \leqslant \sum_{i=0}^{k} \binom{n}{i} \leqslant \frac{(1+t)^n}{t^k} = \frac{(1+\frac{k}{n})^n}{(\frac{k}{n})^k} \leqslant \frac{(e^{\frac{k}{n}})^n}{(\frac{k}{n})^k} = (\frac{en}{k})^k.$$

□

The Inclusion-exclusion Principle (IEP)

Let $A_1,\ A_2,\ \cdots, A_n$ be subsets of Ω (general set). For any $I \subseteq [n]$, denote $A_I := \cap_{i \in I} A_i$. Then $A_\emptyset := \Omega$.

Theorem 3.5.

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_{k=1}^{n} (-1)^{k+1} \sum_{I \in \binom{[n]}{k}} |A_I| = \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} |A_I|.$$

Proof. Rewrite the right hand side

$$\sum_{\emptyset\neq I\subseteq[n]}(-1)^{|I|+1}|A_I| = \sum_{\emptyset\neq I\subseteq[n]}(-1)^{|I|+1}\sum_{x\in A_I}1 = \sum_{x\in\Omega}\sum_{\emptyset\neq I\subseteq[n]:x\in A_I}(-1)^{|I|+1}.$$

Consider the contribution of each $x\in\Omega$ to both sides.

For the left hand side, $|A_1\cup A_2\cup\cdots\cup A_n| = \sum_{x\in\Omega}\delta_x$, where $\delta_x = 1$ if $x\in A_1\cup A_2\cup\cdots\cup A_n$ and 0 otherwise.

For the right hand side, if $x\notin A_1\cup A_2\cup\cdots\cup A_n$, we have

$$\sum_{\emptyset\neq I:x\in A_I}(-1)^{|I|+1} = 0.$$

If $x\in A_1\cup A_2\cup\cdots\cup A_n$, let $J=\{j:\ x\in A_j\}$, then

$$\begin{aligned}\sum_{\emptyset\neq I:x\in A_I}(-1)^{|I|+1} &= \sum_{\emptyset\neq I\subseteq J}(-1)^{|I|+1} = \sum_{i=1}^{|J|}\binom{|J|}{i}(-1)^{i+1}\\ &= (-1)\sum_{i=1}^{|J|}\binom{|J|}{i}(-1)^{i} = (-1)[(1-1)^{|J|}-1] = 1\end{aligned}$$

by the Binomial Theorem. □

Excercise 3.6. *Prove Theorem 3 by induction on n.*

Theorem 3.7. $|A_1^c\cap A_2^c\cap\cdots\cap A_n^c| = \sum_{I\subseteq[n]}(-1)^{|I|}|A_I|$.

Proof.

$$\begin{aligned}&|A_1^c\cap A_2^c\cap\cdots\cap A_n^c| = |\Omega| - |A_1\cup A_2\cup\cdots\cup A_n|\\ =&|A_\emptyset| - \sum_{\emptyset\neq I\subseteq[n]}(-1)^{|I|+1}|A_I| = |A_\emptyset| + \sum_{\emptyset\neq I\subseteq[n]}(-1)^{|I|}|A_I|\\ =&\sum_{I\subseteq[n]}(-1)^{|I|}|A_I|\end{aligned}$$

□

Applications of IEP
Recall.

(1) $S(n,k) = \#$ partitions of $[n]$ into k non-empty parts.

(2) $S(n,k)\cdot k! = \#$ surjections from $[n]$ to $[k]$.

Proposition 3.8. $S(n,k) = \frac{1}{k!}\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n$.

Proof. It suffices to show that the number of surjections from $[n]$ to $[k]$ is

$$\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n.$$

Let $X=[k]$, $Y=[n]$, $\Omega=X^Y$. Define

$$A_i=\{f:\ Y\to X\backslash\{i\}\}$$

for $i\in[k]$, then $|A_i|=(k-1)^n$, and

$$A_I=\cap_{i\in I}A_i=\{f:\ Y\to X\backslash I\}$$

for $I\subseteq[k]$, then $|A_I|=(k-|I|)^n$. Note that # surjections from $[n]$ to $[k]$ is $A_1^c\cap A_2^c\cap\cdots\cap A_k^c$. Using IEP, we can easily get what we want. □

Definition. <u>*Euler number*</u>: $\varphi(n)=\#$ *integers* $m\in[n]$ *s.t.* $\gcd(m,n)=1$.

Proposition 3.9. *If* $n=p_1^{a_1}p_2^{a_2}\cdots p_t^{a_t}$ *where* p_i *are distinct prime divisors of* n*, then* $\varphi(n)=n\prod_{i=1}^{t}(1-\frac{1}{p_i})$.

Proof. Let $\Omega=[n]$, and $A_i=\{m\in[n]:\ p_i|m\}$ for $i\in[t]$. Then $\varphi(n)=|A_1^c\cap\cdots\cap A_t^c|$. Since $|A_i|=\frac{n}{p_i}$ and $|A_I|=\frac{n}{\prod_{i\in I}p_i}$, by IEP,

$$\varphi(n)=\sum_{I\subseteq[t]}(-1)^{|I|}\frac{n}{\prod_{i\in I}p_i}=n\sum_{I\subseteq[t]}(-1)^{|I|}\frac{1}{\prod_{i\in I}p_i}.$$

Note that

$$\prod_{i=1}^{t}(1+x_i)=(1+x_1)\cdots(1+x_t)=\sum_{I\subseteq[t]}\prod_{i\in I}x_i,$$

then

$$\varphi(n)=n\sum_{I\subseteq[t]}(-1)^{|I|}\frac{1}{\prod_{i\in I}p_i}=n\prod_{i=1}^{t}(1-\frac{1}{p_i}).$$

□

Definition. *A permutation* $\sigma:\ X\longrightarrow X$ *is a bijection. It is called a* <u>*derangement*</u> *of* X *if* $\sigma(i)\neq i,\ \forall i\in X$.

Proposition 3.10. *Let* $D_n=\sharp$ *derangement of* $[n]$*, then* $D_n=n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$.

Proof. Let Ω be the set of all permutations of $[n]$, and A_i={all permutations satisfying $\sigma(i) = i$}, $i \in [n]$. Then $|A_i| = (n-1)!$, and $|A_I| = (n - |I|)!$ for each $I \subset [n]$. Then, by IEP,

$$\begin{aligned} D_n =& |A_1^c \cap \cdots \cap A_n^c| = \sum_{I \subseteq [n]} (-1)^{|I|} |A_I| \\ =& \sum_{I \subseteq [n]} (-1)^{|I|} (n - |I|)! = \sum_{i=0}^{n} \binom{n}{i} (-1)^i (n-i)! \\ =& n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}. \end{aligned}$$

□

4 Sep 12: Generating Function

Problem 4.1. *There are 30 red balls, 40 blue balls and 50 white balls in the box, # ways of selecting 70 balls?*

<u>Sol:</u> Consider the coefficient of x^{70} in $(1 + x + \cdots + x^{30})(1 + x + \cdots + x^{40})(1 + x + \cdots + x^{50})$.

Definition. *The (ordinary) <u>Generating function</u> (GF) of an infinity sequence $a_0, a_1, a_2, \cdots$ is a power series $f(x) = \sum_{n \geq 0} a_n x^n$.*

Consequences:
Let $a(x) = \sum_{n \geq 0} a_n x^n$ and $b(x) = \sum_{n \geq 0} b_n x^n$, then

$$a(x) + b(x) = \sum_{n \geq 0} (a_n + b_n) x^n,$$

$$a(x)b(x) = \sum_{n \geq 0} c_n x^n,$$

where $c_n = \sum_{i+j=n} a_i b_j = \sum_{i=0}^{n} a_i b_{n-i}$.

Example 4.2. *(1)* $\frac{1}{1-x} = \sum_{n \geq 0} x^n$ *is the GF of* $\{1, 1, 1, \cdots\}$.

(2) $\frac{1}{1-ax} = \sum_{n \geq 0} (ax)^n = \sum_{n \geq 0} a^n x^n$ *is the GF of* $\{a^0, a^1, \cdots\}$.

(3) $\frac{1}{1-x^2} = \sum_{n \geq 0} x^{2n}$ *is the GF of* $\{1, 0, 1, 0, \cdots\}$.

(4) $f(x) = \sum_{n \geq 0} a_n x^n$ *is the GF of* $\{a_0, a_1, a_2, \cdots\}$.

(5) $cf(x) = \sum_{n\geq 0} ca_n x^n$ *is the GF of* $\{ca_0, ca_1, ca_2, \cdots\}$.

(6) $f(cx) = \sum_{n\geq 0} c^n a_n x^n$ *is the GF of* $\{c^0 a_0, c^1 a_1, c^2 a_2, \cdots\}$.

(7) $f(x^3) = \sum_{n\geq 0} a_n x^{3n}$ *is the GF of* $\{a_0, 0, 0, a_1, 0, 0, \cdots\}$.

(8) $x^3 f(x) = \sum_{n\geq 0} a_n x^{n+3}$ *is the GF of* $\{0, 0, 0, a_0, a_1, \cdots\}$.

(9) $\frac{f(x)-a_0x^0-a_1x^1-a_2x^2}{x^3}$ *is the GF of* $\{a_3, a_4, \cdots\}$.

(10) $f'(x) = \sum_{n\geq 1} na_n x^{n-1}$ *is GF of* $\{a_1, 2a_2, 3a_3, \cdots\}$.

(11) $\sum_{n\geq 1} nx^{n-1} = (\frac{1}{1-x})' = \frac{1}{(1-x)^2}$ *is GF of* $\{1, 2, 3, \cdots\}$.

(12) Take the $(k-1)$*-th derivative of* $\sum_{n\geq 0} x^n = \frac{1}{1-x}$*, we get*

$$\sum_{n\geq k-1} n(n-1)\cdots(n-(k-2))x^{n-(k-1)} = \frac{(k-1)!}{(1-x)^k},$$

then

$$\frac{1}{(1-x)^k} = \sum_{n\geq k-1} \binom{n}{k-1} x^{n-(k-1)} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n.$$

So $\frac{1}{(1-x)^k}$ *is the GF of* $\{a_n\}_{n=0}^{\infty}$*, where* a_n *is the* $\sharp$*integer solutions to* $x_1 + x_2 + \cdots + x_k = n,\ x_i \geq 0$.

Theorem 4.3. *(Generalized binomial theorem) Let* r *be any real number, then* $(1+x)^r = \sum_{n\geq 0} \binom{r}{n} x^n$*, where* $\binom{r}{n} = \frac{(r)(r-1)\cdots(r-n+1)}{n!}$*. We call* $\binom{r}{n}$ *the generalized binomial coefficient.*

Proof. by Taylor expansion. □

Remark. *Since* $(1-x)^{-k} = \sum_{n\geq 0} \binom{-k}{n}(-x)^n$*, and*

$$\begin{aligned}\binom{-k}{n} &= \frac{(-k)(-k-1)\cdots(-k-n+1)}{n!} \\ &= (-1)^n \frac{(n+k-1)\cdots(k+1)k}{n!} = (-1)^n \binom{n+k-1}{n},\end{aligned}$$

then $(1-x)^{-k} = \sum_{n\geq 0} \binom{n+k-1}{n} x^n$*. This is another proof of Example 4.2 (12).*

Ex: Let $a_n + b_1 a_{n-1} + ... + b_k a_{n-k} = 0, n \geq k, b_k \neq 0$ be a k-th linear recursion. The characteristic polynomial of this recursion is defined to be $g(z) = z^k + b_1 z^{k-1} + ... + b_{k-1} z + b_k$. Suppose $g(z)$ has k distinct roots $\lambda_1, ..., \lambda_k$. Then show that $a_n = A_1 \lambda_1^n + ... + A_k \lambda_k^n$, where $A_1, ..., A_k \in \mathbb{C}$.

Recall Problem1: There are 30 red balls, 40 blue balls and 50 white balls in the box, how many ways of selecting 70 balls?

Sol:

$$\begin{aligned}
f(x) &= (1 + x + \cdots + x^{30})(1 + x + \cdots + x^{40})(1 + x + \cdots + x^{50}) \\
&= (\frac{1}{1-x} - \frac{x^{31}}{1-x})(\frac{1}{1-x} - \frac{x^{41}}{1-x})(\frac{1}{1-x} - \frac{x^{51}}{1-x}) \\
&= \frac{(1-x^{31})(1-x^{41})(1-x^{51})}{(1-x)^3} \\
&= (\sum_{n \geq 0} \binom{n+2}{2} x^n)(1-x^{31})(1-x^{41})(1-x^{51}) \\
\Longrightarrow [x^{70}] f(x) &= \binom{72}{2} - \binom{41}{2} - \binom{31}{2} - \binom{21}{2}
\end{aligned}$$

□

Fact: $R(z) = \frac{P(z)}{Q(z)}$ is a rational function, and $\deg(P) < \deg(Q)$. $Q(x) = (z - \lambda_1)^{m_1} ... (z - \lambda_s)^{m_s}$. Then $R(z) = \frac{A_{1,m_1}}{(z-\lambda_1)^{m_1}} + \frac{A_{1,m_1-1}}{(z-\lambda_1)^{m_1-1}} + ... + \frac{A_{1,1}}{z-\lambda_1} + ... + \frac{A_{s,m_s}}{(z-\lambda_s)^{m_s}} + ... + \frac{A_{s,1}}{z-\lambda_s}$.

Problem 4.4. *Fibonacci number: $F_0 = 0, F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for all $n \geq 0$. Compute $\{F_n\}$.*

<u>Sol:</u> Let $f(x) = \sum_{n \geq 0} F_n x^n$, then

$$\begin{aligned}
\sum_{n \geq 0} F_{n+2} x^n &= \sum_{n \geq 0} F_{n+1} x^n + \sum_{n \geq 0} F_n x^n, \\
\Longrightarrow \frac{f(x) - F_0 - F_1 x}{x^2} &= \frac{f(x) - F_0}{x} + f(x), \\
\Longrightarrow f(x) - x &= x f(x) + x^2 f(x), \\
\Longrightarrow f(x) &= \frac{x}{1 - x - x^2} = \frac{a}{1 - \lambda_1 x} + \frac{b}{1 - \lambda_2 x},
\end{aligned}$$

where $\lambda_1 = \frac{1+\sqrt{5}}{2}, \lambda_2 = \frac{1-\sqrt{5}}{2}, a = \frac{1}{\sqrt{5}}, b = -\frac{1}{\sqrt{5}}$. Then

$$\begin{aligned}
f(x) &= \sum_{n \geq 0} a(\lambda_1 x)^n + \sum_{n \geq 0} b(\lambda_2 x)^n = \sum_{n \geq 0} (a\lambda_1^n + b\lambda_2^n) x^n, \\
\Longrightarrow F_n &= \frac{1}{\sqrt{5}} ((\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n)
\end{aligned}$$

□

Remark. *From Problem 4.4,* $\frac{1}{\sqrt{5}}((\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n)$ *is an integer for any* n*, which is close to* $\frac{1}{\sqrt{5}}(\frac{1+\sqrt{5}}{2})^n$ *when* n *goes to infinity.*

5 Sep 17: Catalan Number, GF, EGF

Problem 5.1. $a_n = 2a_{n-1} - a_{n-2}, n \geq 2$. *Compute the GF of* $\{a_n\}$ *and* a_n.

Sol:

$$\sum_{n\geq 2} a_n x^n = \sum_{n\geq 2} 2a_{n-1}x^n - \sum_{n\geq 2} a_{n-2}x^n$$

$$f(x) - a_0 - a_1 x = 2x(f(x) - a_0) - x^2 f(x)$$

$$\begin{aligned}
\Longrightarrow f(x) &= \frac{a_0 + a_1 x}{1 - 2x + x^2} = \frac{a_0 + a_1 x}{(1-x)^2} \\
&= \frac{A}{(1-x)^2} + \frac{B}{1-x} = \frac{a_0 + a_1}{(1-x)^2} + \frac{-a_1}{1-x} \\
&= (a_0 + a_1)\sum_{n\geq 0}\binom{n+1}{1}x^n + (-a_1)\sum_{n\geq 0} x^n \\
&= \sum_{n\geq 0}[(a_0 + a_1)(n+1) - a_1]x^n \\
&\Longrightarrow a_n = (a_0 + a_1)(n+1) - a_1
\end{aligned}$$

□

Problem 5.2. *Let* A_n *be the set of vectors of length* n *with entries from* $\{a, b, c\}$ *and with NO 'aa' occurring. Prove* $a_n = |A_n|, n \geq 1$.

Sol: $a_1 = 3$,$a_2 = 8$. For $n \geq 2$, $a_n = 2a_{n-1} + 2a_{n-2}$. Set $a_0 = 1$, and computing $f(x) = \sum_{n\geq 2} a_n x^n$ is left as exercise. □

Definition. *A* <u>*triangulation*</u> *of an* n*-gon ia a way to add lines between corners to make triangles such that these lines do not cross insides of the* n*-gon. Let* b_{n-1} *be the number of triangulations of an* n*-gon, which is called the* <u>*Catalan number*</u>.

Recall: $\binom{\frac{1}{2}}{n} = \frac{(-1)^{n-1}\cdot 2}{4^n n}\binom{2n-2}{n-1}$.

Problem 5.3. *Compute* b_n.

Sol: $b_2 = 1, b_3 = 2$ and $b_4 = 5$. Denote the vertices of the n-gon by $[n]$, and a triangle with vertices a, b, c by $\Delta_{a,b,c}$. For $4 \le i \le n-1$, $\Delta_{1,2,i}$ can split the n-gon into two parts: an $(i-1)$-gon with vertices $1, 2, \ldots, i$, and an $(n-i+2)$-gon with vertices $1, i, i+1, \ldots, n$. So

$$b_{n-1} = b_{n-2} + \sum_{i=4}^{n-1} b_{i-2} b_{n-i+1} + b_{n-2}.$$

Define $b_0 = 0$ and $b_1 = 1$, then

$$\begin{aligned} b_{n-1} &= b_0 b_{n-1} + b_1 b_{n-2} + \sum_{i=4}^{n-1} b_{i-2} b_{n-i+1} + b_{n-2} b_1 + b_{n-1} b_0 \\ &= \sum_{j=0}^{n-1} b_j b_{n-1-j}, \end{aligned}$$

that is $b_n = \sum\limits_{j=0}^{n} b_j b_{n-j}$ for $n \ge 2$.

Let $f(x) = \sum\limits_{n \ge 0} b_n x^n$, then

$$f^2(x) = \sum_{n \ge 0} (\sum_{j=0}^{n} b_j b_{n-j}) x^n = \sum_{n \ge 2} b_n x^n = f(x) - x.$$

Solving for $f(x)$, we have $f(x) = \frac{1+\sqrt{1-4x}}{2}$ or $f(x) = \frac{1-\sqrt{1-4x}}{2}$. Since $f(0) = 0$, we have

$$\begin{aligned} f(x) &= \frac{1-\sqrt{1-4x}}{2} = \frac{1}{2} - \frac{1}{2}(1-4x)^{\frac{1}{2}} = \frac{1}{2} - \frac{1}{2} \sum_{n \ge 0} \binom{\frac{1}{2}}{n} (-4x)^n \\ &= -\sum_{n \ge 1} \frac{1}{2} \cdot \frac{(-1)^{n+1} 2}{n 4^n} \binom{2n-2}{n-1} (-4)^n x^n = \sum_{n \ge 1} \frac{1}{n} \binom{2n-2}{n-1} x^n, \\ \Longrightarrow b_n &= \frac{1}{n} \binom{2n-2}{n-1}. \end{aligned}$$

□

GF & Selection

Recall:

$$\begin{aligned} \frac{1}{(1-x)^k} &= \sum_{n \ge 0} \binom{n+k-1}{k-1} x^n \\ &= (1 + x + x^2 + \cdots)(1 + x + x^2 + \cdots) \cdots (1 + x + x^2 + \cdots) \\ &= \sum_{n \ge 0} (\sum_{n_1 + n_2 + \cdots + n_k = n, n_i \ge 0} 1) x^n. \end{aligned}$$

Problem 5.4. *Compute a_{20}= #ways to pay 20-yuan given unlimited numbers of three kinds of bills: 1-yuan, 2-yuan, 5-yuan.*

<u>Sol:</u>

$$\begin{aligned} f(x) &= (1+x+x^2+\cdots)(1+x^2+x^4+\cdots)(1+x^5+x^{10}+\cdots) \\ &= \frac{1}{1-x}\frac{1}{1-x^2}\frac{1}{1-x^5}. \end{aligned}$$

Then $[x^{20}]f(x)$ is the answer. □

Integer Partition: write n as a sum of positive integers with no order

Problem 5.5. *Let P_n be the number of the integer partitions of n, find the GF of $\{P_n\}$*

<u>Sol:</u> Let n_j be the number of integer j in such a partition of n, then

$$\sum_{j\geq 1} j\cdot n_j = n.$$

Let $i_j = j\cdot n_j$, which is the contribution of the addends j in a partition of n, then $i_j \in \{0, j, 2j, 3j, \cdots\}$. Let $f_j(x) = 1 + x^j + x^{2j} + x^{3j} + \cdots = \frac{1}{1-x^j}$, then the GF of $\{P_n\}$ is

$$P(x) = \prod_{j\geq 1} f_j(x) = \prod_{j\geq 1}\frac{1}{1-x^j}.$$

□

Remark. *(1) Compute $P_4 = 5$.*

(2) $P_n = e^{\theta(\sqrt{n})}$, i.e. $\exists c_2 \geq c_1 \geq 0$ such that $e^{c_1\sqrt{n}} \leq P_n \leq e^{c_2\sqrt{n}}$.

Recall:

$$\begin{aligned} a_n &= \#\text{integer solutions to } x_1 + ... + x_k = n, x_j \in I_j, j \in [k] \\ &= \sum_{\substack{x1+...+x_k=n \\ x_j\in I_j, j\in[k]}} 1 \end{aligned}$$

Define $f_j(x) = \sum_{i\in I_j} x^i$, then $f(x) = f_1 f_2 ... f_k$ is the generating function of $\{a_n\}$, i.e. $a_n = [x^n]f$.

Recall: Stirling number $S(n,k)$
$z^n = \sum_{k=0}^{n} S(n,k)(z)_k$, where $(z)_k = z(z-1)...(z-k+1)$.
$S(n,k) = kS(n-1,k) + S(n-1,k-1)$, $n \geq 1, k \geq 1$.
$S(n,0) = 0, n > 0$, $S(n,n) = 1, n \geq 0$.

Problem 5.6. *Compute GF of $S(n,k)$ for n. ($k \geq 1$)*

Sol:

$$f_k(x) = \sum_{n\geq 0} S(n,k)x^n$$

.

$$\sum_{n\geq 1} S(n,k)x^n = \sum_{n\geq 1} kS(n-1,k)x^n + \sum_{n\geq 1} S(n-1,k-1)x^n$$

$$f_k(x) = kxf_k(x) + xf_{k-1}(x)$$

$$\Longrightarrow f_k(x) = \frac{x}{1-kx} f_{k-1}(x)$$

By induction on k and $f_0(x) = 1$ we have $f_k(x) = \frac{x^k}{(1-kx)(1-(k-1)x)\ldots(1-x)}$.

$$\Longrightarrow S(n,k) = \sum_{n_1+\ldots+n_k=n-k} k^{n_1}(k-1)^{n_2}\ldots 2^{n_{k-1}}1^{n_k}$$

□

Exponential GF(EGF) and Arrangements

Problem 5.7. *$T_n = \sharp$ ways of picking n balls in order from unlimited number of red, blue and white balls s.t. the $\sharp$ of red and blue balls both even.*

Sol: A selection of e_1 red balls, e_2 blue balls and e_3 white balls with $e_1+e_2+e_3 = n$ contributes $\frac{n!}{e_1!e_2!e_3!}$ to T_n. Therefore

$$T_n = \sum_{\substack{e_1+e_2+e_3=n \\ e_1,e_2\in 2\mathbb{Z}_{\geq 0},\, e_3\in\mathbb{Z}_{\geq 0}}} \frac{n!}{e_1!e_2!e_3!}$$

□

What's the GF of T_n ?

Definition. *The (Exponential) EGF of the sequence $\{a_n\}$ is $f(x) = \sum_{n\geq 0} \frac{a_n}{n!}x^n$.*

Example 5.8. *(1) the EGF of $\{1,1,\cdots\}$ is $\sum_{n\geq 0} \frac{x^n}{n!} = e^x$.*

(2) Let $f_j(x) = \sum_{n\geq 0} \frac{f_j(n)}{n!}x^n$, $j \in [k]$. Then

$$f(x) = \prod_{j=1}^{k} f_j(x) = \prod_{j=1}^{k} (\textstyle\sum_{n\geq 0} \frac{f_j(n)}{n!}x^n) = \sum_{n\geq 0} \left(\sum_{\substack{n_1+\cdots+n_k=n \\ n_1,\cdots n_k\geq 0}} \frac{\Pi_{j=1}^k f_j(n_j)}{n_1!n_2!\cdots n_k!} \right) x^n,$$

i.e. $f(x)$ is the EGF of $a_n = \sum_{\substack{n_1+\cdots+n_k=n \\ n_1,\cdots n_k\geq 0}} \frac{n!\Pi_{j=1}^k f_j(n_j)}{n_1!n_2!\cdots n_k!}$.

(3) Let $f_j(x) = \sum_{n\in I_j} \frac{x^n}{n!}$, $j \in [k]$. Then $f_1 f_2 ... f_k = \sum_{n\geq}(\sum_{\substack{n_1+\cdots+n_k=n \\ n_j\in I_j}} \frac{1}{n_1!n_2!\cdots n_k!})x^n$, i.e. f is the EGF of

$$a_n = \sum^{n_1+\cdots+n_k=n}_{n_j\in I_j} \frac{n!}{n_1!n_2!\cdots n_k!}$$
$$= \#\textit{vectors of length } n \textit{ over } [k], \textit{ s.t. symbol } j \textit{ appears } n_j \textit{ times, where } n_j \in I_j.$$

Recall:

$$k!S(n,k) = \#\text{surjections from } [n] \to [k]$$
$$= \#\text{vertors of length } n \text{ over } [k] \text{ s.t. each symbol appears at least once}$$

Let $I_1 = \cdots = I_k = \{1, 2, \cdots\}$. Let $b_n = k!S(n,k)$. Then the EGF of b_n is

$$\begin{aligned} f(x) = \sum_{n\geq 0} \frac{b_n}{n!} x^n &= \prod_{i=1}^{k} \left(\sum_{n\geq 0} \frac{x^n}{n!} - 1 \right) = (e^x - 1)^k \\ &= \sum_{i=0}^{k} \binom{k}{i} (-1)^{k-i} e^{ix} \\ &= \sum_{n\geq 0} \frac{(ix)^n}{n!} \left(\sum_{i=0}^{k} \binom{k}{i} (-1)^{k-i} \right) \\ &= \sum_{n\geq 0} \frac{\sum_{i=0}^{k} \binom{k}{i} (-1)^{k-i} i^n}{n!} x^n \end{aligned}$$

So $S(n,k) = \frac{b_n}{k!} = \frac{1}{k!} \sum_{i=0}^{k} \binom{k}{i} (-1)^{k-i} i^n$.

Problem 5.9. *Find EGF of $\{T_n\}$ and compute T_n.*

<u>Sol:</u> Let

$$T(x) = \sum_{n=0}^{\infty} \frac{T_n}{n!} x^n,$$

$$f_1(x) = f_2(x) = \sum_{i\in 2\mathbb{Z}_{\geq 0}} \frac{x^i}{i!} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = \frac{e^x + e^{-x}}{2},$$

$$f_3(x) = \sum_{i\in \mathbb{Z}_{\geq 0}} \frac{x^i}{i!} = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

Therefore,

$$T(x) = f_1 f_2 f_3 = \frac{e^{3x} + e^{-x} + 2e^x}{4} = \sum_{n\geq 0} \left(\frac{3^n + 2 + (-1)^n}{4} \right) \frac{x^n}{n!}$$

$\Rightarrow\ T_n = \frac{3^n+2+(-1)^n}{4}$.

$\square$

Problem 5.10. *Let a_n be the number of ways to send n students to 4 classes R_1, R_2, R_3, R_4, s.t. each class has at least one student. Find the EGF of a_n.*

<u>Sol:</u>

$$a_n = \sum_{\substack{i_1+i_2+i_3+n_4=n \\ i_1,\cdots n_4 \geq 1}} \frac{n!}{i_1!i_2!i_3!i_4!}.$$

Let $f_i(x) = \sum_{n\geq 1} \frac{x^n}{n!} = e^x - 1$, $i \in [4]$. Then the EGF of $\{a_n\}$ is

$$f = f_1 f_2 f_3 f_4 = \sum_{n\geq 0} (4^n - 4\cdot 3^n + 6\cdot 2^n - 4)\frac{x^n}{n!} + 1.$$

$\Rightarrow a_n = 4^n - 4\cdot 3^n + 6\cdot 2^n - 4,\ n \geq 1$, and $a_0 = 0$. $\square$

Excercise 5.11. *Use EGF to find $a_n = \#$ vectors of length n over $[k]$.*

Bell number: $B_n = \#$unordered partitions of $[n]$ into disjoint sets. Then $B_n = \sum_{k=1}^{n} S(n,k)$.

Find recurrence first. For B_{n+1}, fix the element $n+1 \in [n+1]$.

If there are exactly k elements in the same subset with $n+1$, then the remaining partitions is B_{n-k}. So

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}, n \geq 0$$

Set $B_0 = 1$. Let $f(x) = \sum_{n\geq 0} \frac{B_n}{n!} x^n$. Then

$$\begin{aligned} f'(x) &= \sum_{n\geq 0} \frac{B_{n+1}}{n!} x^n \\ &= \sum_{n\geq 0}\sum_{k=0}^{n} \frac{x^k}{k!}\frac{B_{n-k}}{(n-k)!} x^{n-k} \\ &= e^x f(x) \end{aligned}$$

$\Longrightarrow f(x) = e^{e^x + C}$. Since $f(0) = B_0 = 1$, $C = -1$.
$\Longrightarrow f(x) = e^{e^x - 1}$.

Hence,

$$
\begin{aligned}
ef(x) = e^{e^x} &= \sum_{k\geq 0} \frac{(e^x)^k}{k!} \\
&= \sum_{k\geq 0} \frac{\left(\sum_{n\geq 0} \frac{(kx)^n}{n!}\right)}{k!} \\
&= \sum_{n\geq 0} \left(\sum_{k\geq 0} \frac{k^n}{k!}\right) \frac{x^n}{n!}. \\
\Longrightarrow B_n &= \frac{1}{e} \sum_{k\geq 0} \frac{k^n}{k!}.
\end{aligned}
$$

6 Sep 19: Dirichlet Convolution, Möbius Inversion

Given $\{a_n\}_{n\geqslant 0}$, the generating function (GF) is $\sum\limits_{n\geqslant 0} a_n x^n$, the exponential generating function (EGF) is $\sum\limits_{n\geqslant 0} \frac{a_n}{n!} x^n$.

Definition. *The* <u>*Dirichlet series*</u> *of* $\{a_n\}_{n=1}^{\infty}$ *is* $a(x) = \sum\limits_{n\geqslant 1} \frac{a_n}{n^x}$.

Let $b(x) = \sum\limits_{n\geqslant 1} \frac{b_n}{n^x}$, $c(x) = a(x)b(x) = \sum\limits_{n\geqslant 1} \frac{c_n}{n^x}$. Then

$$c_n = \sum_{rs=n} a_r b_s = \sum_{d|n} a_d b_{\frac{n}{d}} = \sum_{d|n} a_{\frac{n}{d}} b_d,$$

where $\sum\limits_{d|n}$ means that d runs over all positive divisors of n.

Definition. *The* <u>*Dirichlet Convolution*</u> *of* $f = \{f(n)\}_1^{\infty}$ *and* $g = \{g(n)\}_1^{\infty}$ *is a sequence* $f \odot g$*, where* $(f \odot g)(n) = \sum\limits_{d|n} f(d)g(\frac{n}{d}) = \sum\limits_{d|n} f(\frac{n}{d})g(d)$.

Fact.

(1) $\odot$ is commutative, associative and distributive.

(2) All real sequences form a ring under $\odot$ and $+$.

(3) Let $I = \{I(n)\} = \begin{cases} 1, & n = 1 \\ 0, & n > 1 \end{cases}$. Then $(I \odot f)(n) = \sum\limits_{d|n} I(d)f(\frac{n}{d}) = f(n)$.

(4) If $f \odot g = I$, then we say f is the <u>D-inverse</u> of g, f and g are D-invertible.

(5) Let $e = \{e(n)\}_{n\geqslant 1} = \{1, 1, 1, \cdots\}$. Then $(e \odot f)(n) = \sum\limits_{d|n} e(d)f(\frac{n}{d}) = \sum\limits_{d|n} f(d)$.

Definition. *The* <u>*Möbius Function*</u> $\mu = \{\mu(n)\}_1^\infty$ *is defined by*

$$\mu(n) = \begin{cases} 1 & n = 1, \\ (-1)^r & n = p_1 \cdots p_r \text{ where } p_1, \cdots, p_r \text{ are distinct primes}, \\ 0 & n \text{ is not square-free.} \end{cases}$$

Theorem 6.1. *$\mu \odot e = I$, that is,*

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases}.$$

Proof. The case when $n = 1$ is clearly true.

Suppose $n > 1$. Write $n = p_1^{a_1} \cdots p_r^{a_r}$, where $p_1, \cdots, p_r$ are distinct primes and $a_i \geqslant 1$. If $d \mid n$, then $d = p_1^{b_1} \cdots p_r^{b_r}$ for some $0 \leqslant b_i \leqslant a_i$. If d is not square-free, then $\mu(d) = 0$; If d has no square factors, then $d \mid p_1 \cdots p_r$, which implies $0 \leqslant b_i \leqslant 1$. So

$$\sum_{d|n} \mu(d) = \sum_{d|\prod\limits_{i=1}^{r} p_i} \mu(d) = \sum_{I\subset[r]} \mu(\prod_{i\in I} p_i) = \sum_{I\subset[r]} (-1)^{|I|} = \sum_{i=0}^{r} \binom{r}{i} (-1)^i = 0.$$

□

Lemma 6.2. *$f = \{f(n)\}$ is D-invertible if and only if $f(1) \neq 0$.*

Proof. ($\Rightarrow$) f is D-invertible means $\exists\ g = \{g(n)\}$ such that $f \odot g = I$. So $1 = I(1) = (f \odot g)(1) = f(1)g(1)$, which implies $f(1) \neq 0$.

($\Leftarrow$) We want to prove that $\exists\ g = \{g(n)\}$ such that $f \odot g = I$. We prove it by defining $g(n)$ inductively. For $n = 1$, since $I(1) = f(1)g(1) = 1$, we have $g(1) = \frac{1}{f(1)}$. Assume that we have defined $g(i)$ for all $i < n$, then we define $g(n)$. Since $0 = I(n) = \sum\limits_{d|n, d\neq 1} f(d)g(\frac{n}{d}) + f(1)g(n)$, then $g(n) = -\frac{1}{f(1)} \sum\limits_{d|n, d\neq 1} f(d)g(\frac{n}{d})$. Here, by assumption, all $g(\frac{n}{d})$ are already known. □

Theorem 6.3. *(Möbius Inversion Formula) For any two sequences $\{f(n)\}$ and $\{g(n)\}$, we have*

$$f(n) \equiv \sum_{d|n} g(d) \Longleftrightarrow g(n) \equiv \sum_{d|n} \mu(\frac{n}{d}) f(d)$$

Proof. It is equivalent to prove that $f = e \odot g \Longleftrightarrow g = \mu \odot f$. But this is true left multiplying μ on both sides and the fact that $\mu \odot e = I$ in Theorem 6.1. $\square$

Recall. The Euler function $\varphi(n)$ is defined as the number of integers $m \in [n]$ such that $\gcd(m, n) = 1$. In Proposition 3.9, we have proved that

$$\varphi(n) = \sum_{I \subset [r]} (-1)^{|I|} \frac{n}{\prod\limits_{i \in I} p_i},$$

where n is written as $n = p_1^{k_1} \cdots p_r^{k_r}$. Let $\varphi = \{\varphi(n)\}$.

Theorem 6.4. *Let $N = \{N(n)\} = \{1, 2, 3, \cdots\}$. Then*

(1) $\varphi = N \odot \mu$, i.e. $\varphi(n) = \sum\limits_{d|n} \frac{n}{d} \mu(d)$.

(2) $N = \varphi \odot e$, i.e. $n = \sum\limits_{d|n} \varphi(d)$.

Proof. (1)

$$n \cdot \sum_{d|n} \frac{1}{d} \mu(d) = n \cdot \sum_{d | \prod\limits_{i=1}^{r} p_i} \frac{1}{d} \mu(d) = n \cdot \sum_{I \subset [r]} \frac{1}{\prod\limits_{i \in I} p_i} (-1)^{|I|} = \varphi(n)$$

(2) By (1) and Möbius Inversion Formula.

$\square$

Arrangements in a cycle (without seat number) Here, a cycle of length n over X is $(x_1, x_2, \ldots, x_n)$, where $x_i \in X$, and x_1 is a successor of x_n.

If we don't allow repeated symbols and $|X| = n$, then the number of such cycles is $\frac{n!}{n} = (n-1)!$.

Now we allow repetition and $X = [m]$. Let $C_m(n) = \#$ cycles of length n over $[m]$ with repetition. To find $C_m(n)$, we consider how many lines of length n correspond to the same n-cycle.

Suppose we have an n-cycle of the smallest period p (here $p \mid n$):

$$a_1 a_2 \ldots a_p a_1 a_2 \ldots a_p \ldots a_1 a_2 \ldots a_p.$$

Cut it into lines, then we have p different lines:

$$\begin{array}{ccc} \underline{a_1 a_2 \ldots a_p} & \underline{a_1 a_2 \ldots a_p} \ldots & \underline{a_1 a_2 \ldots a_p} \\ \underline{a_2 \ldots a_p a_1} & \underline{a_2 \ldots a_p a_1} \ldots & \underline{a_2 \ldots a_p a_1} \\ & \vdots & \\ \underline{a_p a_1 \ldots a_{p-1}} & \underline{a_p a_1 \ldots a_{p-1}} \ldots & \underline{a_p a_1 \ldots a_{p-1}} \end{array}$$

Let $L(p) = \#$ lines with period p, $M(p) = \#$ cycles with period p. Then

$$L(p) = pM(p).$$

So we have

$$C_m(n) = \sum_{p|n} M(p) = \sum_{p|n} \frac{1}{p} L(p).$$

Note that $m^n = \sum_{p|n} L(p)$ where m is fixed.

Let $f = \{f(n)\}$ where $f(n) = m^n$, then $f(n) \equiv \sum_{p|n} L(p)$. By Möbius Inversion Formula,

$$L(n) = \sum_{d|n} \mu(\frac{n}{d}) f(d) = \sum_{d|n} \mu(\frac{n}{d}) m^d.$$

So

$$C_m(n) = \sum_{p|n} \frac{1}{p} \sum_{d|p} \mu(\frac{p}{d}) m^d.$$

Excercise 6.5. *Prove that* $C_m(n) = \frac{1}{n} \sum_{d|n} \varphi(\frac{n}{d}) m^d$.

7 Sep 24: Möbius Inversion, Poset

Recall: Let $f = \{f(n)\}$, $g = \{g(n)\}$.

(1) $(f \odot g)(n) = \sum_{d \mid n} f(d) g(\frac{n}{d})$

(2) Identity: $I = \{I(n)\} = \{1, 0, 0, \dots\}$, $f \odot I = I \odot f = f$
if $f \odot g = I$, say f is D-invertible $\Longleftrightarrow f(1) \neq 0$

(3) $e = \{e(n)\} = \{1, 1, 1, \dots\}$, $(f \odot e)(n) = \sum_{d \mid n} f(d)$

$\mu = \{\mu(n)\}$, where $\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^r & n = p_1 p_2 \cdots p_r \\ 0 & n \text{ is not square-free} \end{cases}$

(4) *Möbius Inversion*: $f(n) = \sum_{d|n} g(d) \Longleftrightarrow g(n) = \sum_{d|n} \mu(\frac{n}{d}) f(d)$

(5) Euler Function $\varphi = \{\varphi(n)\}$, and $N = \{N(n)\} = \{1, 2, 3, \dots\}$, then
(i) $N = \varphi \odot e$, i.e. $n = \sum_{d|n} \varphi(d)$;
(ii) $\varphi = N \odot \mu$, i.e. $\varphi(n) = \sum_{d|n} \frac{n}{d} \mu(d)$

(6) $C_m(n) = \sharp$ cycles of length n over $[m]$ with repetition
$L(p) = \sharp$ lines of length n with period p
$M(p) = \sharp$ cycles of length n with period p
$f(n) = m^n = \sharp$ lines of length n over $[m]$

Let's compute $C_m(n)$ in two methods:

(a) $M(p)\cdot p = L(p)$, $C_m(n) = \sum\limits_{d\mid n} M(p)$, $f(n) = \sum\limits_{d\mid n} L(p)$, then

$\Longrightarrow L(n) = \sum\limits_{d|n} \mu(\frac{n}{d}) f(d) \Longrightarrow C_m(n) = \sum\limits_{p|n} \frac{1}{p} \sum\limits_{d|p} \mu(\frac{p}{d}) m^d$

(b) $C = M \odot e$, $f = L \odot e$, $L = NM$,

$\Longrightarrow NC = N(M\odot e) = (NM)\odot(Ne) = L\odot N = f\odot \mu N = f\odot\varphi$

$\Longrightarrow C_m(n) = \frac{1}{n}\sum\limits_{d\mid n} \varphi(\frac{n}{d}) m^d$

Excercise 7.1. *prove $N(f\odot g) = (Nf)\odot(Ng)$ for any two sequences f and g.*

Example 7.2. $C_{10}(9) = \frac{1}{9}(\varphi(9)\cdot 10^1 + \varphi(3)\cdot 10^3 + \varphi(1)\cdot 10^9) = 111111340$

n-th cyclotomic polynomial:
The equation $z^n = 1$ has n complex roots: $z = \theta^0, \theta^1, \cdots, \theta^{n-1}$, where $\theta = e^{\frac{2\pi i}{n}}$. A root θ^k is called primitive if $(k,n) = 1$. The n-th cyclotomic polynomial is defined as

$$\Phi_n(z) = \prod_{\substack{1\le i\le n\\ (i,n)=1}} (z-\theta^i).$$

Theorem 7.3. $z^n - 1 = \prod\limits_{d|n} \Phi_d(z)$

Proof. Note that $z^n - 1 = \prod\limits_{t=0}^{n-1}(z-\theta^t)$. If $(t,n) = d$, then $(\frac{t}{d}, \frac{n}{d}) = 1$. Let $t' = \frac{t}{d}, n' = \frac{n}{d}$, then $\theta^t = e^{\frac{2\pi i}{n}t} = e^{\frac{2\pi i}{n'}t'}$. So θ^t is n'-th primitive root.

$$\Longrightarrow z^n - 1 = \prod_{d|n} \prod_{\substack{t=0\\(t,n)=d}}^{n-1} (z-\theta^t) = \prod_{d|n} \Phi_{\frac{n}{d}}(z) = \prod_{d|n}\Phi_d(z)$$

□

Theorem 7.4. $\Phi_n(z) = \prod\limits_{d|n} (z^d - 1)^{\mu(\frac{n}{d})}$

Proof. Note that $\ln(z^n - 1) = \sum\limits_{d|n} \ln(\Phi_d(z))$. By inversion,

$\Longrightarrow \ln(\Phi_n(z)) = \sum\limits_{d|n} \mu(\frac{n}{d}) \ln(z^d - 1)) \Longrightarrow \Phi_n(z) = \prod\limits_{d|n} (z^d-1)^{\mu(\frac{n}{d})}$. □

Inversion Formula on Poset

Definition. *A partially ordered set(**poset**) $P = (X, \le)$ is a set X with a relation "$\le$" on X, s.t.*

(1) Reflectivity: $x \le x$;

(2) Antisymmetry: If $x \le y$ and $y \le x$, then $x = y$;

(3) Transitivity: If $x \leq y$ and $y \leq z$, then $x \leq z$.

Example 7.5. *(1) $(\mathbb{Z}_{\geq 0}, <)$, with usual order.*

(2) $(\mathbb{Z}_{>0}, \leq)$, divisibility relation, $a \leq b \Leftrightarrow a \mid b$.

(3) $(2^X, \leq)$, inclusion relation, $A \leq B \Leftrightarrow A \subseteq B$.

Definition. *A poset is called locally finite if $[a, b] = \{z : a \leq z \leq b\}$ is finite for any a, b.*

Definition. *$P = (X, \geq)$, locally finite, the incidence algebra of P is*

$$\mathbb{A}(P) = \{f : P^2 \to \mathbb{R} | f(x, y) = 0, whenever\ x \not\leq y\}$$

Example 7.6. *(1) $0(x, y) = 0$.*

(2) delta function: $\delta(x, y) = \begin{cases} 1, & x = y \\ 0, & x \neq y \end{cases}$.

(3) Zeta function: $\zeta(x, y) = \begin{cases} 1, & x \leq y \\ 0, & else \end{cases}$.

Facts:

(1) $f, g \in \mathbb{A}(P) \Rightarrow f + g \in \mathbb{A}(P)$.

(2) $f \in \mathbb{A}(P) \Rightarrow cf \in \mathbb{A}(P), \forall c \in \mathbb{R}$.

Definition. *Let $f, g \in \mathbb{A}(P)$, the <u>Dedekind convolution</u> of f and g is $f * g \in \mathbb{A}(P)$, where $(f * g)(x, y) = \sum\limits_{z\,:\,x \leq z \leq y} f(x, z) g(z, y)$.*

Fact:

(1) $*$ is non-commutative, associative, distributive.

(2) $\mathbb{A}(P), *, +, \cdot$ form an incidence algebra.

(3) $(f * 0)(x, y) = 0$.
$(f * \delta)(x, y) = f(x, y)\delta(y, y) = f(x, y) = (\delta * f)(x, y)$, so δ is the identity.
$(f * \zeta)(x, y) = \sum\limits_{z\,:\,x \leq z \leq y} f(x, z)$

Definition. *If $f * g = \delta$, we say f is left-inverse of g.*

Remark. *if $f * g_1 = \delta = g_2 * f$, then $g_1 = g_2$.*

Theorem 7.7. *$f \in \mathbb{A}(P)$, then f has a (left)right inverse $\Longleftrightarrow f(x, x) \neq 0$, $\forall x \in P$.*

Proof. ($\Rightarrow$) $\exists g \in \mathbb{A}(P)$, s.t. $f * g = \delta \implies \delta(x,x) = f(x,x)g(x,x) = 1 \implies f(x,x) \neq 0$

($\Leftarrow$) Our aim is to find $g \in \mathbb{A}(P)$, s.t. $f * g = \delta$. We define g inductively as in Lemma 6.2.

$$\begin{cases} f(x,x)g(x,x) = \delta(x,x) = 1 & \implies \quad g(x,x) = \frac{1}{f(x,x)}, \forall x \\ \sum\limits_{\substack{z \\ x \leq z \leq y}} f(x,z)g(z,y) = \delta(x,y) = 0 & \quad x \neq y \end{cases}.$$

Then

$$f(x,x)g(x,y) + \sum_{\substack{z \\ x < z \leq y}} f(x,z)g(z,y) = 0,$$

i.e. $g(x,y) = -\frac{1}{f(x,x)} \left(\sum\limits_{\substack{z \\ x < z \leq y}} f(x,z)g(z,y) \right) = 0$, by recursion. $\square$

Remark. *The left-inverse g of f is* $g(x,y) = -\frac{1}{f(y,y)} \left(\sum\limits_{\substack{z \\ x \leq z < y}} g(x,z)f(z,y) \right)$.

Definition. *Möbius Function over P is $\mu_P = \zeta^{-1}$, where*

$$\mu_P(x,y) = \begin{cases} 1, & x = y \\ -\sum\limits_{x < z \leq y} \mu_P(z,y) = -\sum\limits_{x \leq z < y} \mu_P(x,z), & x < y \\ 0, & else \end{cases}.$$

Theorem 7.8. ***(Inverse Formula I)*** *Suppose that the poset P has a* <u>*minimum*</u> *element m (i.e. $m \leq x$, $\forall x \in P$). Let f, g be functions $P \to \mathbb{R}$,then*

$$f(y) = \sum_{z \leq y} g(z), \ \forall y \in P \iff g(y) = \sum_{z \leq y} f(z)\mu_P(z,y), \ \forall y \in P$$

Remark. *m' is a* <u>*minimal*</u> *element of P if $\nexists x \in P$ s.t. $x \leq m'$*

8 Sep 26: Inversion Formula, Möbius function

Recall.

$\mathbb{A}(P) = \{f : P \times P \to \mathbb{R} | f(x,y) = 0 \text{ whenever } x \nleqslant y\}$

$f * g$ is defined as $(f * g)(x,y) = \sum\limits_{z : x \leqslant z \leqslant y} f(x,z)g(z,y)$ for $x \leqslant y$

identity: $\delta(x,y) = \begin{cases} 1, & x = y \\ 0, & x \neq y \end{cases}$.

Zeta function: $\zeta(x,y) = \begin{cases} 1, & x \leqslant y \\ 0, & x \nleqslant y \end{cases}$.

Möbius function: $\mu_P = \zeta^{-1}$ where

$$\mu_P(x,y) = \begin{cases} 1, & x = y \\ -\sum\limits_{x<z\leqslant y} \mu_P(z,y) = -\sum\limits_{x\leqslant z<y} \mu_P(x,z), & x < y \\ 0, & else \end{cases}$$

Theorem 8.1. *(Inversion Formula I) Suppose P has a minimum element m (i.e $m \leqslant x,\ \forall\ x \in P$). Let f, g be functions $P \to \mathbb{R}$, then*

$$f(y) = \sum_{z\leqslant y} g(z),\ \forall\ y \in P \Longleftrightarrow g(y) = \sum_{z\leqslant y} f(z)\mu_P(z,y),\ \forall\ y \in P.$$

Proof. Define $\overline{f}(x,y) = \begin{cases} f(y), & x = m \\ 0, & x \neq m \end{cases}$, $\overline{g}(x,y) = \begin{cases} g(y), & x = m \\ 0, & x \neq m \end{cases}$. Then $\overline{f}, \overline{g} \in \mathbb{A}(P)$.

Claim: $\overline{f} = \overline{g} * \zeta$.

Proof of Claim: If $x = m$, then

$$\overline{f}(m,y) = f(y) = \sum_{z\leqslant y} g(z) = \sum_{z:m\leqslant z\leqslant y} \overline{g}(m,z)\zeta(z,y) = (\overline{g} * \zeta)(m,y).$$

If $x > m$, then

$$\overline{f}(x,y) = 0 = \sum_{z:x\leqslant z\leqslant y} \overline{g}(x,z)\zeta(z,y) = (\overline{g} * \zeta)(x,y).$$

So $\overline{g} = \overline{f} * \mu_P$, which implies

$$g(y) = \overline{g}(m,y) = \sum_{z:m\leqslant z\leqslant y} \overline{f}(m,z)\mu_P(z,y) = \sum_{z\leqslant y} f(z)\mu_P(z,y).$$

□

Theorem 8.2. *(Inversion Formula II) Suppose P has a maximum element M (i.e $x \leqslant M,\ \forall\ x \in P$). Let f, g be functions $P \to \mathbb{R}$, then*

$$f(x) = \sum_{x\leqslant z} g(z),\ \forall\ x \in P \Longleftrightarrow g(x) = \sum_{x\leqslant z} \mu_P(x,z)f(z),\ \forall\ x \in P.$$

Proof. Similar to the proof above !

Define $\overline{f}(x,y) = \begin{cases} f(x), & y = M \\ 0, & y \neq M \end{cases}$, $\overline{g}(x,y) = \begin{cases} g(x), & y = M \\ 0, & y \neq M \end{cases}$.

Claim: $\overline{f} = \zeta * \overline{g}$.

So $\overline{g} = \mu_P * \overline{f}$, which implies

$$g(x) = \overline{g}(x,M) = \sum_{z:x\leqslant z\leqslant M} \mu_P(x,z)\overline{f}(z,M) = \sum_{x\leqslant z} \mu_P(x,z)f(z).$$

□

Example 8.3. *Suppose that $P = C_n = \{x_1 < \cdots < x_n\}$ is a totally ordered set of size n. $\forall\ f \in \mathbb{A}(P)$, define A_f by $A_f(i,j) = f(x_i, x_j)$, then A_f is an upper triangle matrix. Claim: $A_{f*g} = A_f \cdot A_g$.*

Proof of Claim:

$$\begin{aligned} A_{f*g}(i,j) = (f*g)(x_i,x_j) &= \sum_{x_i \leqslant z \leqslant x_j} f(x_i,z)g(z,x_j) \\ &= \sum_{z \in P} f(x_i,z)g(z,x_j) = \sum_{k=1}^{n} f(x_i,x_k)g(x_k,x_j) \\ &= \sum_{k=1}^{n} A_f(i,k)A_g(k,j) \end{aligned}$$

□

Definition. *$P = (X, \leqslant)$, $|X| = n$, a <u>linear extension</u> of P is permutation $x_1x_2\cdots x_n$ such that $x_i < x_j$ implies $i < j$.*

Fact. Any finite poset has a linear extension.

Lemma 8.4. *Let μ_P be the Möbius function of P.*

(1) If $\varphi : P \to Q$ is an isomorphism (i.e. $x \leqslant y \Leftrightarrow \varphi(x) \leqslant \varphi(y)$), then $\mu_P(x,y) = \mu_Q(\varphi(x), \varphi(y))$.

(2) $u, v \in P$, $P_1 = [u,v]$, then $\mu_{P_1}(x,y) = \mu_P(x,y)$ for $x, y \in [u,v]$.

(3) If P is a direct product of $(P_i, \leqslant_i)$ for $i \in [k]$, i.e. $P = P_1 \times P_2 \times \cdots \times P_k$ and $(x_1, x_2, \cdots, x_k) \leqslant (y_1, y_2, \cdots, y_k) \Leftrightarrow x_i \leqslant_i y_i$ for $i \in [k]$, then $\mu_P(x,y) = \prod_{i=1}^{k} \mu_{P_i}(x_i, y_i)$.

Example 8.5. *Continue of Example 8.3. By Claim, we have $A_{\mu_{C_n}} = A_{\zeta^{-1}} = (A_\zeta)^{-1}$. So*

$$\mu_{C_n}(x,y) = \begin{cases} 1, & x = y \\ -1, & y \text{ cover } x,\ \text{i.e.}[x,y] = \{x,y\} for\ x < y \\ 0, & else \end{cases}$$

Let $P = (\mathbb{Z}_{\geqslant 0}, \leqslant)$ with usual order. Then Lemma 8.4 implies

$$\mu_P(x,y) = \mu_{[1,n]}(x,y) = \mu_{C_n}(\varphi(x), \varphi(y)).$$

So

$$\mu_P(x,y) = \begin{cases} 1, & x = y \\ -1, & y = x+1 \\ 0, & else \end{cases}$$

Then $f(y) = \sum_{z \leqslant y} g(z) \Longleftrightarrow g(y) = \sum_{z \leqslant y} f(z)\mu_P(z,y) = f(y) - f(y-1)$.

Example 8.6. *$\Omega = (\mathbb{Z}_{>0}, \leqslant)$ is the divisor poset, i.e. $a \leqslant b \Leftrightarrow a \mid b$, then $\mu_\Omega(x,y) = \mu(\frac{y}{x})$ if $x \mid y$.*

Proof. If $x \mid y$, consider the isomorphism φ from poset $[x,y]$ to $[1, \frac{y}{x}]$ by $\varphi(z) = \frac{z}{x}$. By Lemma 8.4, we have

$$\mu_\Omega(x,y) = \mu_{[x,y]}(x,y) = \mu_{[1,\frac{y}{x}]}(1, \frac{y}{x}) \triangleq f(\frac{y}{x}).$$

It suffices to show that $f(n) = \mu(n)$ for $\forall\ n > 0$.

If $n = 1$, then $\mu(1) = 1 = f(1)$. If $n > 1$, let $n = p_1^{e_1} \cdots p_k^{e_k}$. Then $[1,n] = \{p_1^{h_1} \cdots p_k^{h_k} : 0 \leqslant h_i \leqslant e_i\}$. Define $\overline{\Omega} = \{(h_1, \cdots, h_k) : 0 \leqslant h_i \leqslant e_i\}$ to be a poset with

$$(h_1, \cdots, h_k) \leqslant (t_1, \cdots, t_k) \Leftrightarrow h_i \leqslant t_i \text{ for } \forall\ i \in [k].$$

So

$$(h_1, \cdots, h_k) \leqslant (t_1, \cdots, t_k) \Leftrightarrow p_1^{h_1} \cdots p_k^{h_k} \mid p_1^{t_1} \cdots p_k^{t_k},$$

that is, $[1,n]$ is isomorphic to $\overline{\Omega}$. For each $i \in [k]$, let $\Omega_i = \{0, 1, \cdots, e_i\}$ with usual order. Then $\overline{\Omega} = \Omega_1 \times \cdots \times \Omega_k$. By Lemma 8.4, we have

$$\begin{aligned} f(n) &= \mu_{[1,n]}(1,n) = \mu_{\overline{\Omega}}((0,\cdots,0),(e_1,\cdots,e_k)) = \prod_{i=1}^{k} \mu_{\Omega_i}(0,e_i) \\ &= \begin{cases} (-1)^k & e_1 = \cdots = e_k = 1 \\ 0, & e_i \geqslant 2 \text{ for some } i \end{cases} \end{aligned}$$

which implies $f(n) = \mu(n)$. □

9 Oct: Inversion Formula on Poset, Group Action, Burnside Lemma

Theorem 9.1. ***(Inverse Formula I)*** *If P has a minimum element m(i.e. $m \leq x$, $\forall x \in P$). Let f, g be functions $P \to \mathbb{R}$,then*

$$f(y) = \sum_{z \leq y} g(z),\ \forall y \in P \iff g(y) = \sum_{z \leq y} f(z)\mu_P(z,y),\ \forall y \in P$$

Theorem 9.2. ***(Inverse Formula II)*** *If P has a maximum element M(i.e. $x \leq M$, $\forall x \in P$). Let f, g be functions $P \to \mathbb{R}$,then*

$$f(x) = \sum_{x \leq z} g(z),\ \forall x \in P \iff g(x) = \sum_{x \leq z} \mu_P(x,z)f(z),\ \forall x \in P$$

Example 9.3. *$P = (2^X, \subseteq)$, then $\mu_P(A,B) = (-1)^{|B|-|A|}$.*

Proof. Let $X = [n], \varphi : 2^X \longrightarrow \{0,1\}^n, A \longmapsto \varphi(A)$, where $\varphi(A)$ is the indicator vector. Let $S = \{0,1\}, S^n = \{0,1\}^n = S \times S \times \cdots \times S$, define

$$(x_1, x_2, \ldots, x_n) \leq (y_1, y_2, \ldots, y_n) \Leftrightarrow x_i \leq y_i, \forall i \in [n],$$

then $A \subseteq B \Leftrightarrow \varphi(A) \leq \varphi(B)$, φ is isomorphism. So

$$\begin{aligned}\mu_P(A,B) &= \mu_{S^n}(\varphi(A), \varphi(B)) = \mu_{S^n}((x_1, x_2, \ldots, x_n), (y_1, y_2, \ldots, y_n)) \\ &= \prod_{i=1}^{n} \mu_S(x_i, y_i) = \prod_{i=1}^{n} (-1)^{y_i - x_i} = (-1)^{|B|-|A|}\end{aligned}$$

□

Theorem 9.4. ***(IEP)*** $|A_1^c \cap \cdots \cap A_n^c| = \sum\limits_{I \subseteq [n]} (-1)^{|I|} |A_I|$.

Proof. Let $X = [n], P = (2^X, \subseteq)$.

Define

$$f: \ P \to \mathbb{R} \text{ as } f(|I|) = |A_I| = |\bigcap_{i \in I} A_i|,$$

$$g: \ P \to \mathbb{R} \text{ as } g(|I|) = |A_I \cap (\bigcap_{j \notin I} A_j^c)|.$$

Then $f(I) = \sum\limits_{I \subseteq J \subseteq [n]} g(J)$. Since P has a maximal element $[n]$, by *Inverse Formula II*

$$g(I) = \sum_{I \subseteq J \subseteq [n]} \mu_P(I,J) f(J) = \sum_{I \subseteq J \subseteq [n]} (-1)^{|J \setminus I|} |A_J|.$$

Particularly, $g(\emptyset) = |A_1^c \cap \cdots \cap A_n^c| = \sum\limits_{I \subseteq [n]} (-1)^{|I|} |A_I|$.

□

Polya Counting Theorem

Definition. *a finite group $(G, \cdot)$, finite set X, a* <u>*group action*</u> *(G, X) is a map $* : G \times X \to X$,s.t.*

(1) $e * x = x$,where e is id of $G, \forall x \in X$.

(2) $\forall g, h \in G, \forall x \in X, g * (h * x) = (g \cdot h) * x$.

Example 9.5. *(1) $(S_n, \circ), X = [n]$, define a group action by $\sigma * (i) = \sigma(i)$.*

(2) $(G, \cdot), X = G$, define a group action by $g * h = g \cdot h, \forall g, h \in G$.

Definition. *(1) Orbit of $x, Orb(x) = \{g * x : \forall g \in G\}$.*

(2) Stabilizer of $x, Stab(x) = \{g \in G : g * x = x\}$.

(3) Fixed points of $g \in G, Fix(g) = \{x \in X : g * x = x\}$.

Theorem 9.6. *(Orbit-Stabilizer Thm)* $|G| = |Orb(x)| \cdot |Stab(x)|$.

Proof. Let $G_x = Stab(x)$, and $G = g_1G_x \cup g_2G_x \cup \cdots \cup g_nG_x$. Denote $G/G_x = \{g_iG_x : i \in [n]\}$, then we have $|G| = n|G_x| = |G/G_x| \cdot |G_x|$. Define a function $f : G/G_x \to Orb(x)$ by $f(g_iG_x) = g_i * x$.

Ex: check f is a bijection and complete the proof. □

Theorem 9.7. *(Burnside Lemma) Given a finite group action* (G, X). *Let* $N(G)$ *be number of distinct orbits of* X, *then* $N(G) = \frac{1}{|G|}\sum_{g\in G} |Fix(g)|$.

Proof. (Double Counting) Consider the set $S = \{(g, x) : g * x = x\}$.

(i) group (g, x) by the first element

$$|S| = \sum_{g\in G} |\{x : g * x = x\}| = \sum_{g\in G} |Fix(g)|.$$

(ii) group (g, x) by the second element

$$\begin{aligned} |S| &= \sum_{x\in X} |\{g : g * x = x\}| = \sum_{x\in X} |Stab(x)| \\ &= \sum_{i=1}^{N(G)} \sum_{x\in Orb(x_i)} |Stab(x_i)| = \sum_{i=1}^{N(G)} |Orb(x_i)| \cdot |Stab(x_i)| \\ &= N(G) \cdot |G|. \end{aligned}$$

□

Example 9.8. $\tau = (123)(45)$, $G =< \tau >$, $X = [5]$, *then* $Orb(1) = Orb(2) = Orb(3) = \{1, 2, 3\}$, $Orb(4) = Orb(5) = \{4, 5\}$ *and* $2 = N(G) = \frac{1}{6}(5 + 0 + 2 + 3 + 2) = 2$

Theorem 9.9. *(Weighted Burnside Lemma) Finite group action* (G, X). *Let* $\Omega_i = Orb(x_i), i \in [N(G)]$ *be all distinct orbits. Let* $\omega : X \to R$ *be a weighted function, where* $\omega(x) = \omega(y)$ *if* $x, y \in \Omega_i$, *for some* $i \in [N(G)]$. *Define* $\omega(\Omega_i) = \omega(x_i)$, *then*

$$\sum_{i=1}^{N(G)} \omega(\Omega_i) = \frac{1}{|G|} \sum_{g\in G} \sum_{x\in Fix(g)} \omega(x).$$

Proof. Let $S = \{(g, x) : g * x = x\}$. Then applying double counting as in Theorem 9.7 to compute $M = \sum_{(g,x)\in S} \omega(x)$.

(i)

$$M = \sum_{g \in G} \sum_{x \in Fix(g)} \omega(x).$$

(ii)

$$\begin{aligned} M &= \sum_{x \in X} \sum_{g \in Stab(x)} \omega(x) = \sum_{x \in X} \omega(x)|Stab(x)| \\ &= \sum_{i=1}^{N(G)} \sum_{x \in \Omega_i} \omega(x)|Stab(x)| = \sum_{i=1}^{N(G)} \omega(x_i)|\Omega_i||Stab(x_i)| \\ &= |G| \sum_{i=1}^{N(G)} \omega(\Omega_i). \end{aligned}$$

□

Definition. *A* <u>*coloring*</u> *of X in m colors is a function $f : X \to C$, where C is a set of m colors.*

Let C^X ={all coloring of X in C}. Given a group action (G, X), we define a induced group action (G, C^X): for each $g \in G$ and $f \in C^X$, define $g * f \in C^X$ as

$$(g * f)(x) = f(g^{-1} * x)$$

10 Oct: Polya Theorem

Recall. (Burnside Lemma) For any finite group action (G, X). Let $N(G)$ be the number of distinct orbits of X, then $N(G) = \frac{1}{|G|} \sum_{g \in G} |Fix(g)|$.

Definition. *(1) A coloring of X in m colors is a function $f : X \to C$, where C is a set of m colors.*

(2) C^X ={all colorings of X in C}.

(3) (G, X) induces a group action (G, C^X) as $\forall\ g \in G,\ \forall\ f \in C^X$, define $g * f$ by $(g * f)(x) = f(g^{-1} * x)$, $\forall\ x \in X$.

Fact. Any finite group can be viewed as a permutation group.

Example 10.1. *(1) $X = [6]$, $g = (1\ 2\ 3\ 4\ 5\ 6)$, $C = \{g, r, b\}$*
$f_1 : 1 \to r \quad 2 \to b \quad 3 \to b \quad 4 \to g \quad 5 \to r \quad 6 \to b$
$f_2 = g * f_1 : 1 \to b \quad 2 \to r \quad 3 \to b \quad 4 \to b \quad 5 \to g \quad 6 \to r$

(2) $X = [6]$, $g = (1\ 3\ 5)(2\ 4\ 6)$, $C = \{g, r, b\}$
$f_1 : 1 \to r \quad 2 \to b \quad 3 \to b \quad 4 \to g \quad 5 \to r \quad 6 \to b$
$f_2 = g * f_1 : 1 \to r \quad 2 \to b \quad 3 \to r \quad 4 \to b \quad 5 \to b \quad 6 \to g$

Remark. *$f \in C^X$ is fixed by g if and only if f has the same color along each cycle of g, so $|Fix(g)| = |C|^{c(g)}$ where $c(g)$ is the number of cycles of g.*

Theorem 10.2. *(Polya Theorem) $|X| = n$, $|C| = m$, $G \leqslant S_n$. Then,*

$$\#\textit{different colorings in } C^X \textit{ under } G = \#\textit{orbits of } C^X = \frac{1}{|G|}\sum_{g\in G} m^{c(g)}.$$

Definition. *For any $g \in S_n$, let $\lambda_i(g)$ be the number of cycles of length i in G. Then the cycle type of g is type$(g) = (\lambda_1(g), \lambda_2(g), \cdots, \lambda_n(g))$.*

Example 10.3. *$g = (135)(246)$, type$(g) = (0, 0, 2, 0, 0, 0)$.*

Definition. *$G \leqslant S_n$, the <u>cycle index</u> of G is a polynomial*

$$P_G(x_1, x_2, \cdots, x_n) = \frac{1}{|G|}\sum_{g\in G} x_1^{\lambda_1(g)} x_2^{\lambda_2(g)} \cdots x_n^{\lambda_n(g)}.$$

Corollary 10.4. *(Polya Theorem)*

$$\frac{1}{|G|}\sum_{g\in G} m^{c(g)} = P_G(m, m, \cdots, m).$$

Problem 10.5. *Let $\sigma_n =< (1, 2, \cdots, n) >$. Compute $P_{\sigma_n}(x_1, x_2, \cdots, x_n)$.*

Solution. Let $\tau = (1, 2, \cdots, n)$, then $\sigma_n = \{\tau, \tau^2, \cdots, \tau^n\}$.

For $\forall\ k \in [n]$, consider τ^k. What is *type*(τ^k)? $\forall\ i \in [n], \tau^k(i) \equiv i + k \pmod n$. Suppose the cycle containing i has length l, then the cycle is $(i, i + k, \cdots, i + (l - 1)k)$. So $l = \dfrac{n}{(k, n)}$. So, all cycles of τ^k have length $\dfrac{n}{(k, n)}$, i.e.

$$\lambda_l(\tau^k) = \begin{cases} 0 & l \neq \dfrac{n}{(k,n)}; \\ (k, n) & l = \dfrac{n}{(k,n)}. \end{cases}$$

So,

$$\begin{aligned} P_{\sigma_n}(x_1, x_2, \cdots, x_n) &= \frac{1}{n}\sum_{i=1}^{n} (x_{\frac{n}{(k,n)}})^{(k,n)} \\ &= \frac{1}{n}\sum_{j|n}\ \sum_{k:(k,n)=j} (x_{\frac{n}{j}})^j \end{aligned}$$

Let

$$\begin{aligned} r(j) &= \#\{k \in [n] :\ (k, n) = j\} \\ &= \#\{\frac{k}{j} \in [\frac{n}{j}] : (\frac{k}{j}, \frac{n}{j}) = 1\} \\ &= \varphi(\frac{n}{j}). \end{aligned}$$

So,

$$P_{\sigma_n}(x_1, x_2, \cdots, x_n) = \frac{1}{n}\sum_{j|n}\varphi(\frac{n}{j})(x_{\frac{n}{j}})^j$$
$$= \frac{1}{n}\sum_{d|n}\varphi(d)(x_d)^{\frac{n}{d}}.$$

Example 10.6. *$C_m(n) = \#$ orbits of $[m]^{[n]}$ under $\sigma_n = P_{\sigma_n}(m, m, \cdots, m) = \frac{1}{n}\sum_{d|n}\varphi(d)m^{\frac{n}{d}}$.*

Definition. *The dihedral group $D_n \leqslant S_n$ is*

$$D_n = \{\tau, \tau^2, \cdots, \tau^n\} \cup \{\pi_1, \pi_2, \cdots, \pi_n\},$$

where $\pi_k(i) \equiv -i + k \pmod n$.

Problem 10.7. *Show that*

$$P_{D_n}(x_1, x_2, \cdots, x_n) = \begin{cases} \frac{1}{2}P_{\sigma_n}(x_1, x_2, \cdots, x_n) + \frac{1}{2}x_1 x_2^{\frac{n-1}{2}} & n \text{ odd;} \\ \frac{1}{2}P_{\sigma_n}(x_1, x_2, \cdots, x_n) + \frac{1}{4}(x_2^{\frac{n}{2}} + x_1^2 x_2^{\frac{n-2}{2}}) & n \text{ even.} \end{cases}$$

Solution.

$$P_{D_n}(x_1, x_2, \cdots, x_n) = \frac{1}{2n}(\sum_{i=1}^{n} x_1^{\lambda_1(\tau^i)} \cdots x_n^{\lambda_n(\tau^i)} + \sum_{i=1}^{n} x_1^{\lambda_1(\pi_i)} \cdots x_n^{\lambda_n(\pi_i)})$$
$$= \frac{1}{2}P_{\sigma_n}(x_1, \cdots, x_n) + \frac{1}{2n}\sum_{i=1}^{n} x_1^{\lambda_1(\pi_i)} \cdots x_n^{\lambda_n(\pi_i)}$$

Since π_k is a reflection, i.e. $(\pi_k)^2 = id$, π_k is a product of m_k cycles of length 1 and $\frac{n-m_i}{2}$ cycles of length 2.
So

$$\frac{1}{2n}\sum_{i=1}^{n} x_1^{\lambda_1(\pi_i)} \cdots x_n^{\lambda_n(\pi_i)} = \frac{1}{2n}\sum_{i=1}^{n} x_1^{m_i} x_2^{\frac{n-m_i}{2}}.$$

$m_k = |Fix(\pi_k)| = \#\{i : -i + k \equiv i \pmod n\} = \#\{i : 2i \equiv k \pmod n\}$.
If n is odd, $(2, n) = 1$, 2 has inverse in $\mathbb{Z}_n$, so $m_1 = \cdots = m_n = 1$. So,

$$\frac{1}{2n}\sum_{i=1}^{n} x_1^1 x_2^{\frac{n-1}{2}} = \frac{1}{2}x_1 x_2^{\frac{n-1}{2}}.$$

If n is even, then $2i \equiv k \pmod n$ has no solutions if k is odd, and two solutions $i = \frac{k}{2}$ and $i = \frac{k}{2} + \frac{n}{2}$ if k is even.
So

$$m_k = \begin{cases} 0 & k \text{ odd;} \\ 2 & k \text{ even.} \end{cases}$$

Then,

$$\frac{1}{2n}\sum_{i=1}^{n} x_1^{m_k} x_2^{\frac{n-m_k}{2}} = \frac{1}{2n}(\frac{n}{2}x_1^0 x_2^{\frac{n}{2}} + \frac{n}{2}x_1^2 x_2^{\frac{n-2}{2}})$$
$$= \frac{1}{4}(x_2^{\frac{n}{2}} + x_1^2 x_2^{\frac{n-2}{2}}).$$

□

Corollary 10.8. *# cycles of length $[n]$ over $[m]$ under rotations and reflections is*

$$P_{D_n}(m, \cdots m) = \frac{1}{2}C_m(n) + \begin{cases} \frac{1}{2}m^{\frac{n+1}{2}} & n \text{ odd}; \\ \frac{1}{4}(m^{\frac{n}{2}} + m^{\frac{n}{2}+1}) & n \text{ even}. \end{cases}$$

Definition. *(1) Simple graph $H = (V, E)$. V: vertex set, $E \subseteq \binom{V}{2}$: edge set.*

(2) If $\{i, j\} \in E$, then we say $\{i, j\}$ is incident with i and j, and i is adjacent to j.

(3) $H = (V, E)$, $\varphi : V \to V$. If $\forall\ u, v \in V$, $\{u, v\} \in E \Leftrightarrow \{\varphi(u), \varphi(v) \in E\}$, then we say φ is an automorphism of H.

(4) $Aut(H) = \{$all automorphisms of $H\} \leqslant Sym(V)$.

Example 10.9. *(1) K_n: complete graph. $Aut(K_n) = S_n$.*

(2) $C_n = (V, E)$: a cycle of length n. $Aut(C_n) = D_n$.

11 Oct: Polya Theorem

Recall (Polya Theorem) $|X| = n$, $|C| = m$, $G \leqslant S_n$. Then,

$$\#\text{ orbits in } C^X \text{ under } G = P_G(m, \cdots, m)$$

where $P_G(x_1, x_2, \cdots, x_n) = \frac{1}{|G|}\sum_{g\in G} x_1^{\lambda_1(g)} x_2^{\lambda_2(g)} \cdots x_n^{\lambda_n(g)}$.

Problem 11.1. *Consider coloring of a regular cube in m colors under rotations in 3-dimension space.*

Solution. Let G be the group of automorphisms including all rotations. Now we look at nontrivial elements of G. There are three pairs of opposite faces. For each pair of opposite faces, you can rotate the cube along the axis through the two centers by an angle of 90°, 180° and 270°. So there are $3 \times 3 = 9$ such rotations (type I rotations). Similarly, there are six pairs of opposite edges, for each pair, there is exactly one rotation along the axis through the two centers of opposite edges by 180° (type II rotations). There are four pairs of opposite

vertices, and for each pair, there are two rotations by 120° and 240° (type III rotations). So in total, there are $3 \times 3 + 6 \times 1 + 4 \times 2 = 23$ nontrivial rotations in G. Including the identity element, we have $|G| = 24$.

Note that the group of rotations of the cube G may induce different automorphism groups of the set of vertices, the set of edges and the set of faces.

(1) Consider colorings of vertices $X = \{x_1, \cdots, x_8\}$, let G_1 be the induced automorphism group on vertices. Based on the above analysis of different kinds of rotations, G_1 has

   (a) one identity element with cycle type $(8, 0, 0, 0, 0, 0, 0, 0)$;

   (b) type I rotations: six elements each consist of four cycles of length two, i.e., cycle type $(0, 4, 0, 0, 0, 0, 0, 0)$; three elements each consist of two cycles of length four, i.e., cycle type $(0, 0, 0, 2, 0, 0, 0, 0)$;

   (c) type II rotations: six elements each consist of four cycles of length two, i.e., cycle type $(0, 4, 0, 0, 0, 0, 0, 0)$;

   (d) type III rotations: eight element each consist of two cycles of length one and two cycles of length three, i.e., cycle type $(2, 0, 2, 0, 0, 0, 0,)$.

   So, $P_{G_1}(x_1, \cdots, x_8) = \dfrac{1}{24}({x_1}^8 + 3(2{x_4}^2 + {x_2}^4) + 6{x_2}^4 + 2{x_1}^2{x_3}^2 \times 4)$,
   which implies $P_{G_1}(m, \cdots, m) = \dfrac{1}{24}(m^8 + 17m^4 + 6m^2)$.

(2) Consider colorings of faces $X = \{x_1, \cdots, x_6\}$, let G_2 be the induced automorphism group on faces. Then G_2 has

   (a) one identity element with cycle type $(6, 0, 0, 0, 0, 0)$;

   (b) type I rotations: six elements each consist of two cycles of length one and one cycle of length four, i.e., cycle type $(2, 0, 0, 1, 0, 0)$; three elements each consist of two cycles of length one and two cycles of length two, i.e., cycle type $(2, 2, 0, 0, 0, 0)$;

   (c) type II rotations: six elements each consist of three cycles of length two, i.e., cycle type $(0, 3, 0, 0, 0, 0)$;

   (d) type III rotations: eight element each consist of two cycles of length three, i.e., cycle type $(0, 0, 2, 0, 0, 0)$.

   So, $P_{G_2}(x_1, \cdots, x_6) = \dfrac{1}{24}({x_1}^6 + 3(2{x_1}^2x_4 + {x_1}^2{x_2}^2) + 6{x_2}^3 + 8{x_3}^2)$,
   which implies $P_{G_2}(m, \cdots, m) = \dfrac{1}{24}(m^6 + 3m^4 + 12m^3 + 8m^2)$.

(3) **(Exercise.)** Consider colorings of edges $X = \{x_1, \cdots, x_{12}\}$

□

Problem 11.2. *Let H be the following graph, $G = Aut(H)$, $X = \{A, B, C, D, E, F\}$, $|C| = m$. Consider C^X.*

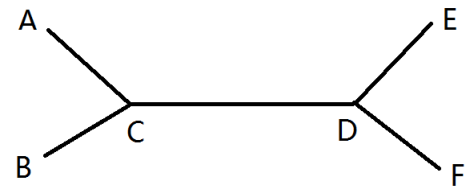

Figure 1: H

Solution.

(1) By considering the automorphisms of the graph H, we have $|Aut(H)| \geqslant 8$. (enumeration)

(2) $|Stab(A)| = 2$, $|Orbit(A)| = 4$, so $|G| = 8$.

Now it's easy to calculate $P_G(x_1, \cdots, x_6)$ by Polya Theorem. $\square$

Recall (Weighted Burnside Lemma) Finite group action (G, X). Let $\Omega_i = Orb(x_i)$, $i \in [N(G)]$ be all distinct orbits. Let $\omega : X \to \mathbb{R}$ be a weighted function, where $\omega(x) = \omega(y)$ if $x, y \in \Omega_i$ for some $i \in [N(G)]$. Define $\omega(\Omega_i) = \omega(x_i)$, then

$$\sum_{i=1}^{N(G)} \omega(\Omega_i) = \frac{1}{|G|} \sum_{g \in G} \sum_{x \in Fig(x)} \omega(x).$$

Problem 11.3. *The number of cycles of length n over $[m]$ such that i occurs k_i times, $i \in [m]$.*

Definition. *Given a group action (G, X) and a set of colorings C^X. Let $\omega : C \longrightarrow \mathbb{R}$ be any function, then ω induces a weight function of C^X as follows:*

$$\forall\ f \in C^X, \textit{define } \omega(f) = \prod_{x \in X} \omega(f(x)).$$

If $f_1,\ f_2 \in C^X$ are in the same orbit, i.e. $f_2 = g * f_1$, i.e. $f_2(x) = f_1(g^{-1} * x)$, then $\omega(f_2) = \prod_{x \in X} \omega(f_2(x)) = \prod_{x \in X} \omega(f_1(g^{-1} * x)) = \prod_{x \in X} \omega(f_1(x)) = \omega(f_1)$. i.e. ω has the same value on each orbit.

Theorem 11.4. *(G, C^X), $\omega : C^X \longrightarrow \mathbb{R}$ is defined as above. $\mathcal{F} = \{$all orbits in $C^X\}$. Then*

$$\sum_{F \in \mathcal{F}} \omega(F) = P_G(\sum_{c \in C} \omega(c), \sum_{c \in C} \omega(c)^2, \cdots, \sum_{c \in C} \omega(c)^n).$$

Corollary 11.5. *(G, C^X), $|x| = n$, $C = \{c_1, c_2, \cdots, c_m\}$, $k_1+k_2+\cdots+k_m = n$, $k_i \geq 0$. Let $N(k_1, \cdots k_m)$ be the number of colorings in Problem 11.3. Then*

$$P_G(\sum_{i=1}^{m} y_i, \sum_{i=1}^{m} y_i^2, \cdots, \sum_{i=1}^{m} y_i^n) = \sum_{k_1+\cdots+k_m=n} N(k_1, \cdots, k_m) y_1^{k_1} \cdots y_m^{k_m}.$$

Proof. Let $\omega(c_i) = y_i$, $i \in [m]$. Let f be a coloring with c_i occuring k_i times, $i \in [m]$. Then

$$\omega(f) = \prod_{x \in X} \omega(f(x)) = y_1^{k_1} \cdots y_m^{k_m}.$$

$$\begin{aligned} N(k_1, \cdots, k_m) &= \# \text{ orbits } F \text{ such that } \omega(F) = y_1^{k_1} \cdots y_m^{k_m} \\ &= \text{coefficient of } y_1^{k_1} \cdots y_m^{k_m} \text{ in } \sum_{F \in \mathcal{F}} \omega(F) \\ &= \text{coefficient of } y_1^{k_1} \cdots y_m^{k_m} \text{ in } P_G(\sum_{i=1}^{m} y_i, \sum_{i=1}^{m} y_i^2, \cdots, \sum_{i=1}^{m} y_i^n) \end{aligned}$$

□

Solution to Problem 11.3 Let $G = \sigma_n$. $P_{\sigma_n}(x_1, \cdots, x_n) = \frac{1}{n} \sum_{d|n} \varphi(d)(x_d)^{\frac{n}{d}}$. Then

$$P_{\sigma_n}(\sum_{i=1}^{m} y_i, \sum_{i=1}^{m} y_i^2, \cdots, \sum_{i=1}^{m} y_i^n) = \frac{1}{n} \sum_{d|n} \varphi(d)(\sum_{i=1}^{m} y_i^d)^{\frac{n}{d}}.$$

Let $T(k_1, \cdots, k_m)$ be the coefficient of $y_1^{k_1} \cdots y_m^{k_m}$. If $d \nmid k_i$, then no monomial $y_1^{k_1} \cdots y_m^{k_m}$. So

$$T(k_1, \cdots, k_m) = \frac{1}{n} \sum_{d|(k_1, \cdots, k_m)} \varphi(d) \frac{(\frac{n}{d})!}{(\frac{k_1}{d})! \cdots (\frac{k_m}{d})!}.$$

□

12 Oct: Polya Thm, SDR

Recall:

1. $N(k_1, \cdots k_m) = \sharp$ colorings of C^X under G s.t. there are k_i elements of X have color c_i, $\sum_{i \in [m]} k_i = |X| = n$
2. Weight Function: define $\omega : C \longrightarrow \mathbb{R}$, $\forall f \in C^X$, $\omega(f) = \prod_{x \in X} \omega(f(x))$
3. Weighted Burnside Lemma: $\sum_{i \in [N(G)]} \omega(\Omega_i) = \frac{1}{|G|} \sum_{g \in G} \sum_{x \in Fix(g)} \omega(x)$

Theorem 12.1. *(G, C^X), $\omega : C^X \longrightarrow \mathbb{R}$ as defined above. $\mathcal{F} = \{$all orbits in $C^X\}$, then*

$$\sum_{F\in\mathcal{F}} \omega(F) = P_G(\sum_{c\in C}\omega(c), \sum_{c\in C}\omega(c)^2, \cdots, \sum_{c\in C}\omega(c)^n).$$

Proof. $\forall g \in G$, let $k = k(g)$ be the number of cycles of g and $A_1, \cdots, A_k$ be the set of elements of each cycle. For $f \in Fix(g)$, suppose $f(x) = c_i$, if $x \in A_i, i \in [k]$. Then $\omega(f) = \prod_{i=1}^{k}(\omega(c_i)^{|A_i|})$. So

$$\sum_{f\in Fix(g)} \omega(f) = \sum_{(c_1,\cdots,c_k)\in C^k} (\prod_{i=1}^{k}(\omega(c_i)^{|A_i|})) = \prod_{i=1}^{k}(\sum_{c\in C}\omega(c)^{|A_i|}).$$

Denote $type(g) = (\lambda_1(g), \cdots, \lambda_n(g))$, then

$$\sum_{f\in Fix(g)} \omega(f) = \prod_{i=1}^{n}(\sum_{c\in C}\omega(c)^i)^{\lambda_i(g)}$$

By weighted Burnside Lemma,

$$\begin{aligned}\sum_{F\in\mathcal{F}} \omega(F) &= \frac{1}{|G|}\sum_{g\in G}\sum_{f\in Fix(g)}\omega(f)\\ &= \frac{1}{|G|}\sum_{g\in G}(\prod_{i=1}^{n}(\sum_{c\in C}\omega(c)^i)^{\lambda_i(g)})\\ &= P_G(\sum_{c\in C}\omega(c), \sum_{c\in C}\omega(c)^2, \cdots, \sum_{c\in C}\omega(c)^n).\end{aligned}$$

□

Corollary 12.2.

$$P_G(\sum_{i=1}^{m} y_i, \sum_{i=1}^{m} y_i^2, \cdots, \sum_{i=1}^{m} y_i^n) = \sum_{k_1+\cdots+k_m=n} N(k_1, \cdots, k_m) y_1^{k_1}\cdots y_m^{k_m}.$$

Definition. *$(G, X), (H, Y)$, where $X \cap Y = \emptyset$. Let $G \times H = \{(g, h) : g \in G, h \in H\}$, define $(g, h) : X \cup Y \longrightarrow X \cup Y$ by*

$$(g, h) * (a) = \begin{cases} g * a, & if\ a \in X \\ h * a, & if\ a \in Y. \end{cases}$$

Then $G \times H$ is permutation group of $X \cup Y$.

Lemma 12.3. *$(G \times H, X \cup Y), |X| = n, |Y| = m$. Then*

$$P_{G\times H}(x_1, \cdots, x_n, y_1, \cdots, y_m) = P_G(x_1, \cdots, x_n)P_H(y_1, \cdots, y_m).$$

Proof. Exercise. $\square$

<u>Distribution Problem</u>:

n balls with r colors, n_i balls with color i, $\sum_{i=1}^{r} n_i = n$, put them into m boxes: $B_1, \cdots, B_m$, s.t. there are k_i balls in B_j, where $\sum_{i=1}^{m} k_i = n$.

(1) If $n_i = 1, i \in [r]$ and $r = n$, then it is an ordered partition problem, s.t. the $i-$th part is of size $k_i, i \in [m]$, so $\frac{n!}{k_1! \cdots k_m!}$.

(2) In general, it is a coloring problem as follows. $X_1, \cdots, X_r$, where $|X_i| = n_i$ with permutation group G_i, since each ball in X_i are considered to be the same, $G_i = S_{n_i}$. So we have a group action $S_{n_1} \times \cdots \times S_{n_r}$ over $X_1 \cup \cdots X_r$. By lemma 1, it has cycle index $\prod\limits_{i=1}^{r} P_{s_i}(x_1, \cdots, x_{n_i})$. then total number of distributions is $\prod\limits_{i=1}^{r} P_{s_{n_i}}(m, \cdots, m)$. So

$$N(k_1, \cdots, k_n) = [y_1^{k_1} \cdots y_m^{k_m}] \prod_{i=1}^{n} P_{S_{n_i}}(\sum_{l=1}^{m} y_l, \cdots, \sum_{l=1}^{m} y_l^{n_i}).$$

(3) If $r = 1$, all balls have same color, $n_1 = n$. Then the total number of distributions is the number of solutions to $k_1 + \cdots + k_m = n, k_i \geq 0$, that is $\binom{n+m-1}{m-1} = \binom{n+m-1}{n}$. Since the cycle index is $P_{s_n}(x_1, \cdots, x_n)$, we have $P_{s_n}(m, \cdots, m) = \binom{n+m-1}{n}$. Hence

$$\prod_{i=1}^{r} P_{s_{n_i}}(m, \cdots, m) = \prod_{i=1}^{r} \binom{n_i + m - 1}{n_i}.$$

(4) $r = n, \quad n_i = 1$, the number of n different balls into m numbered boxes.

$$\prod_{i=1}^{n} P_{s_1}(m) = \prod_{i=1}^{n} m = m^n.$$

Then

$$\begin{aligned} N(k_1, \cdots, k_m) =& \frac{n!}{k_1! \cdots k_m!} = [y_1^{k_1} \cdots y_m^{k_m}](\prod_{i=1}^{n} P_{s_1}(\sum_{i=1}^{m} y_i)) \\ =& [y_1^{k_1} \cdots y_m^{k_m}](\sum_{i=1}^{m} y_i)^n \end{aligned}$$

(5) The number of putting a, a, b, b into boxes A, B? Here $r = 2, n_1 = n_2 = 2$.

$$\prod_{i=1}^{2} P_{s_{n_i}}(2, 2) = \binom{2+2-1}{2}^2 = 9.$$

(6) The number of putting a, a, b, b, c, c into boxes A, B, C. s.t. 3 balls in A, 2 balls in B, 1 ball in C?

$$P_{s_2}(x_1, x_2) = \frac{1}{2}(x_1^2 + x_2).$$

$$P_{s_2}(y_1 + y_2 + y_3, y_1^2 + y_2^2 + y_3^2) = \frac{1}{2}[(y_1 + y_2 + y_3)^2 + y_1^2 + y_2^2 + y_3^2].$$

$$[y_1^3 y_2^2 y_3][P_{s_2}(y_1 + y_2 + y_3, y_1^2 + y_2^2 + y_3^2)]^3 = 15.$$

Remark. *(1) numbered balls and numbered boxes: vectors*

(2) unnumbered balls and numbered boxes: #solutions to $\sum_{i=1}^{m} k_i = n$

(3) numbered balls and unnumbered boxes: stirling number

(4) unnumbered balls and unnumbered boxes: integer partition

SDR

Definition. *A system of distinct representatives(SDR) for a sequence of set* $S_1, S_2, \cdots, S_m$*(are not necessarily distinct) is a sequence of distinct elements* $x_1, \cdots, x_m$*, s.t.* $x_i \in S_i, i \in [m]$.

Theorem 12.4. ***(Hall's marriage Theorem)*** *The sets of* $S_1, S_2, \cdots, S_m$ *has a SDR iff* $|\cup_{i\in I} S_i| \geq |I|$ *for* $I \subset [m]$*(Hall's condition) i.e. for* $\forall k \in [m]$*, the union of any* k *sets has at least* k *elements.*

Proof. "$\Longrightarrow$" If $S_1, S_2, \cdots, S_m$ have a SDR $x_1, \cdots, x_m$, then $\forall I \subset [m], |\cup_{i\in I} S_i| \geq |\{x_i, i \in I\}| = |I|$.

$\Longleftarrow$ Prove by induction on m. The case $m = 1$ is clear. Assume the claim holds for any collection with $< m$ sets. We split it into two cases.

(1) If for all $I \subsetneq [m], |\cup_{i\in I} S_i| > |I|$, take $x \in S_1$ as its representative, let $S_i' = S_i \backslash \{x\}, i = 2, \cdots, m$. then for all $I \subset [2, m], |\cup_{i\in I} S_i'| \geq |I|$. by assumption $S_2', \cdots, S_m'$ have a SDR $x_2, \cdots, x_m$. then $x_1, \cdots, x_m$ is a SDR of $S_1, \cdots, S_m$.

(2) If for some $I \subsetneq [m], |\cup_{i\in I} S_i| = |I| = k$ for some k. by recording $S_i, i \in [m]$, we may assume $|\cup_{i=1}^{k} S_i| = k$. by assumption, $S_1, \cdots, S_k$ have a SDR $x_1, \cdots, x_k$. Let $S_i' = S_i \backslash \{x_1, \cdots, x_k\}, i \in [k+1, m]$. then for all $I \subset [k+1, m], |\cup_{i\in I} S_i'| \geq |I|$, if not, $|(\cup_{i\in I} S_i) \cup (\cup_{j=1}^{k} S_j)| < |I| + k$, a contradiction. So by assumption $S_{k+1}', S_{k+2}', \cdots, S_m'$ have a SDR $x_{k+1}, \cdots, x_m$, then $x_1, x_2, \cdots, x_m$ is a SDR of $S_1, S_2, \cdots, S_m$. $\square$

Corollary 12.5. $|X| = n$, $|S_i| = r, S_i \subset X, i \in [m].s.t.|\{i : x \in S_i\}| = d$ *for all* $x \in X$*. If* $m \leq n$*, then* $S_1, \cdots, S_m$ *have a SDR.*

Proof. Consider the incidence matrix $M = (m_{x,i})$ of $S_i, i \in [m]$. That is M is a 0-1 matrix with $|X|$ rows labeled by elements $x \in X$, and with m column labeled by $i \in [m]$, such that $m_{x,i} = 1$ iff $x \in S_i$, count the number of 1 in M. we have $dn = mr$, then $m \leq n$ implies $d \leq r$. Suppose $S_1, \cdots, S_m$ don't have a SDR, by Hall's Theorem, $\exists k$ sets $S_{i_1}, \cdots, S_{i_k}$ for some $k \in [m]$,

$$|Y| = |S_{i_1} \cup \cdots \cup S_{i_k}| < k$$

$\forall x \in Y$, let $d_x = |\{j \in [k] : x \in S_{i_j}\}| \leq d$. count the number of 1 in rows labeled by $x \in Y$ and columns $i_j, j \in [k]$.

$$rk = \sum_{j=1}^{k} |S_{i_j}| = \sum_{x \in Y} d_x \leq d|Y| < dk.$$

a contradiction.

□